

Active and Passive Components used in Power Electronics

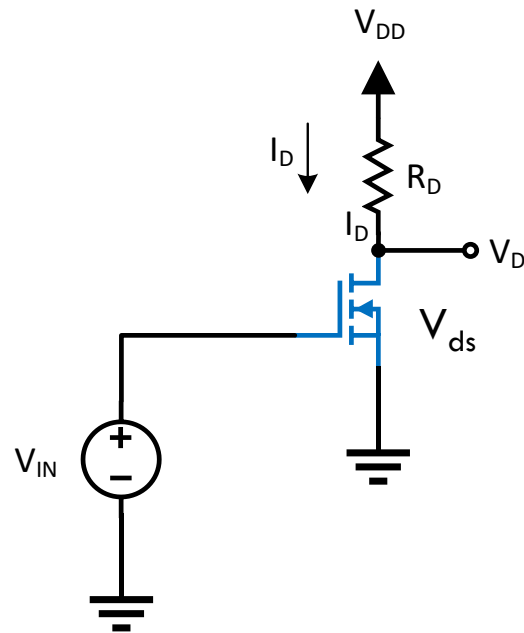
BASICS OF MOSFET POWER LOSS

Power losses (P_t) in any component operating in the switch-mode can be divided in three groups:

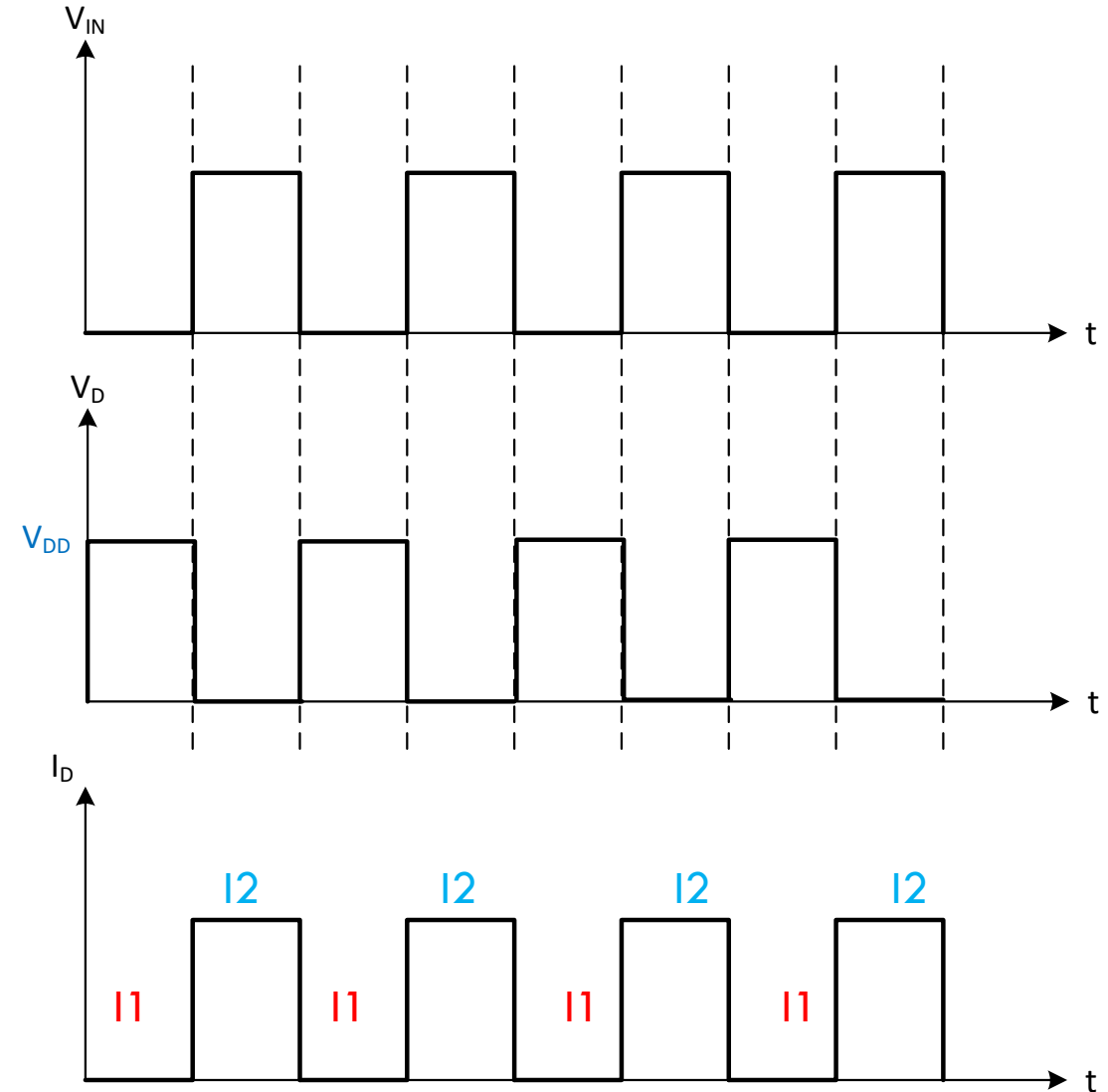
a) Conduction losses (P_{cond})

b) Switching losses (P_{sw})

$$P_t = P_{\text{cond}} + P_{\text{sw}}$$



Ideally $I_1 = 0$ A, however a real MOSFET has leakage current, therefore $I_1 \neq 0$ A

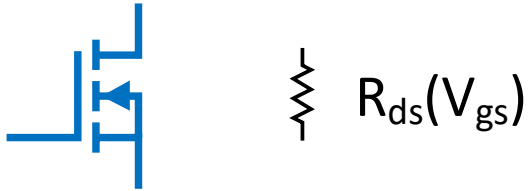


BASICS OF MOSFET POWER LOSS

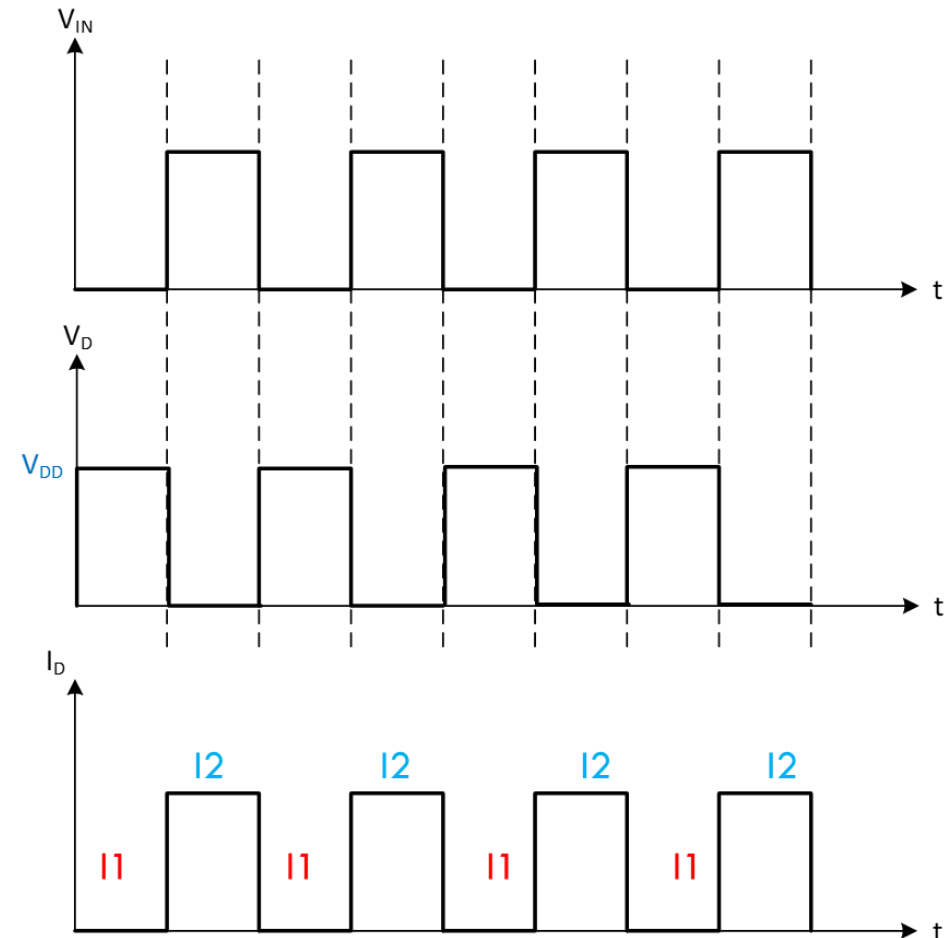
Conduction Loss:

$$P_{\text{cond}}(t) = I_d^2(t) \cdot R_{ds}$$

$$P_{\text{cond}} = \frac{1}{T_{\text{sw}}} \int_0^{T_{\text{sw}}} P_{\text{cond}}(t) dt = \frac{1}{T_{\text{sw}}} \int_0^{T_{\text{sw}}} R_{ds}(t) I_d^2(t) dt = R_{ds} I_{d,\text{rms}}^2$$

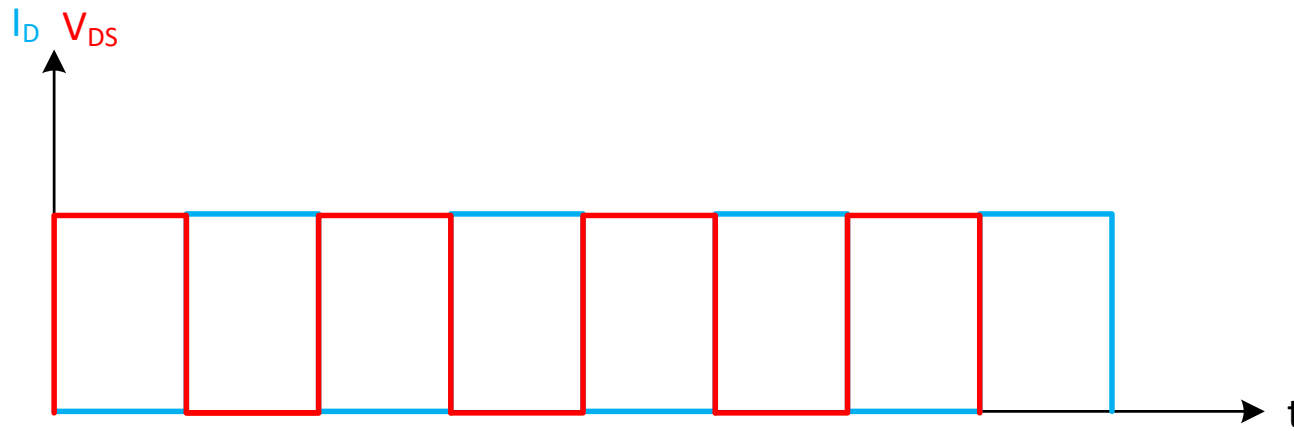


In linear region, the MOSFET behaves like a switch and can be represented with a resistor which is dependent on gate-source voltage



BASICS OF MOSFET POWER LOSS

Switching Loss:

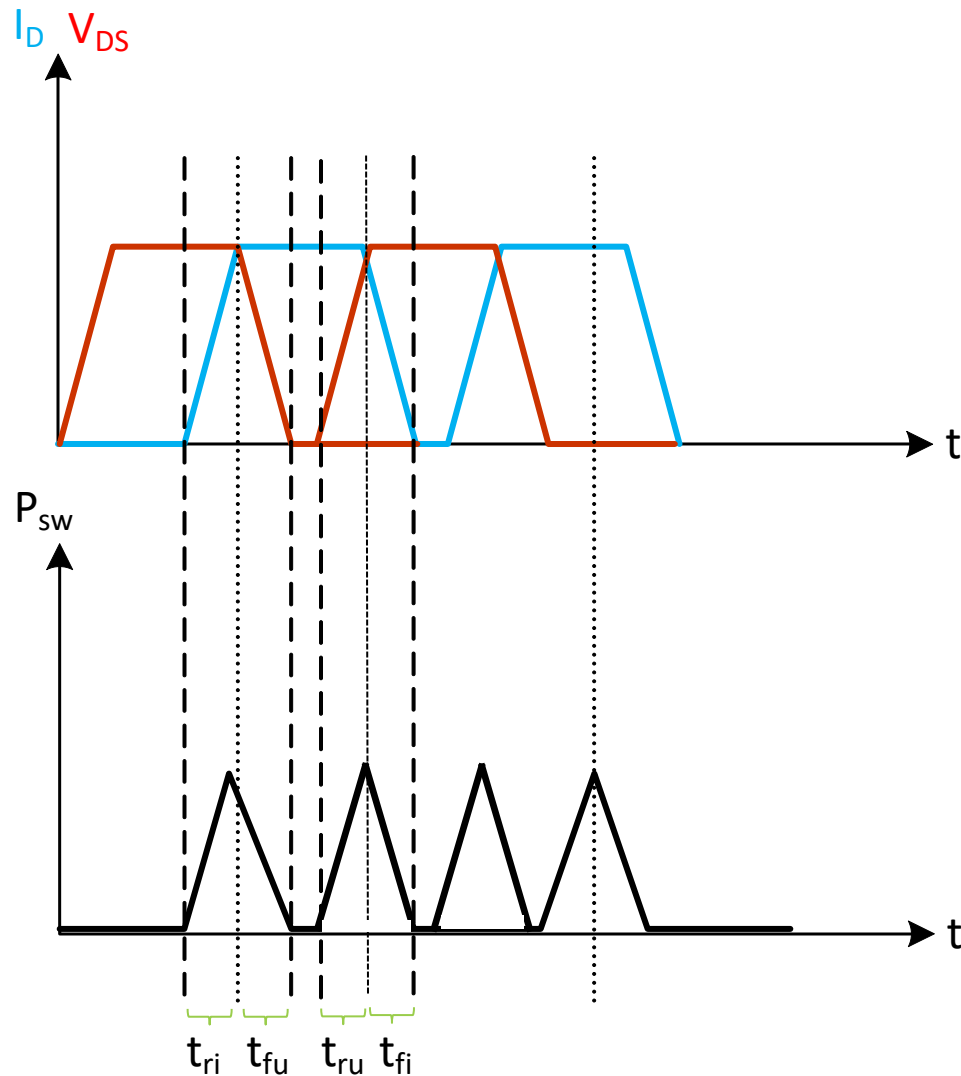


Ideal switch, no switching losses

However, the rise and fall times of a real MOSFET is not zero which leads to an overlap of drain-source voltage and drain current

BASICS OF MOSFET POWER LOSS

Switching Loss: for an ideal MOSFET, t_{ri} , t_{fu} , t_{ru} and t_{fi} are zero, for a real MOSFET, the switching losses are calculated as follows:



Switching Energies:

$$E_{on} \approx \int_0^{t_{ri}+t_{fu}} V_{DS}(t) I_D(t) dt = \frac{t_{ri}+t_{fu}}{2} V_D I_D$$

$$E_{off} \approx \int_0^{t_{ru}+t_{fi}} V_{DS}(t) I_D(t) dt = \frac{t_{ru}+t_{fi}}{2} V_D I_D$$

$$P_{sw} = (E_{on} + E_{off}) f_{sw}$$

UNDERSTANDING THE DATASHEET OF TRANSISTORS

MOSFET

Table 6. Dynamic characteristics

List of constants and limitations relating to the table i.e. voltages, currents and temperatures

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$Q_{G(tot)}$	total gate charge	$I_D = 20\text{ A}; V_{DS} = 44\text{ V}; V_{GS} = 10\text{ V}$	-	35.2	-	nC
Q_{GS}	gate source charge		-	9.24	-	nC
Q_{GD}	gate drain charge		-	14.8	-	nC
C_{iss}	input capacitance	$V_{DS} = 25\text{ V}; V_{GS} = 0\text{ V}; f = 1\text{ MHz}; T_j = 25\text{ }^\circ\text{C}$	-	1550	2067	pF
C_{oss}	output capacitance		-	328	394	pF
C_{rss}	reverse transfer capacitance		-	153	210	pF
$t_{d(on)}$	turn-on delay time	$V_{DS} = 30\text{ V}; V_{GS} = 10\text{ V}; R_L = 1.5\text{ }\Omega;$	-	19.3	-	ns
t_r	rise time	$R_{G(ext)} = 10\text{ }\Omega$	-	29.4	-	ns
$t_{d(off)}$	turn-off delay time		-	43.2	-	ns
t_f	fall time		-	22	-	ns

Rise and Fall times for a predefined R_G Gate Resistor

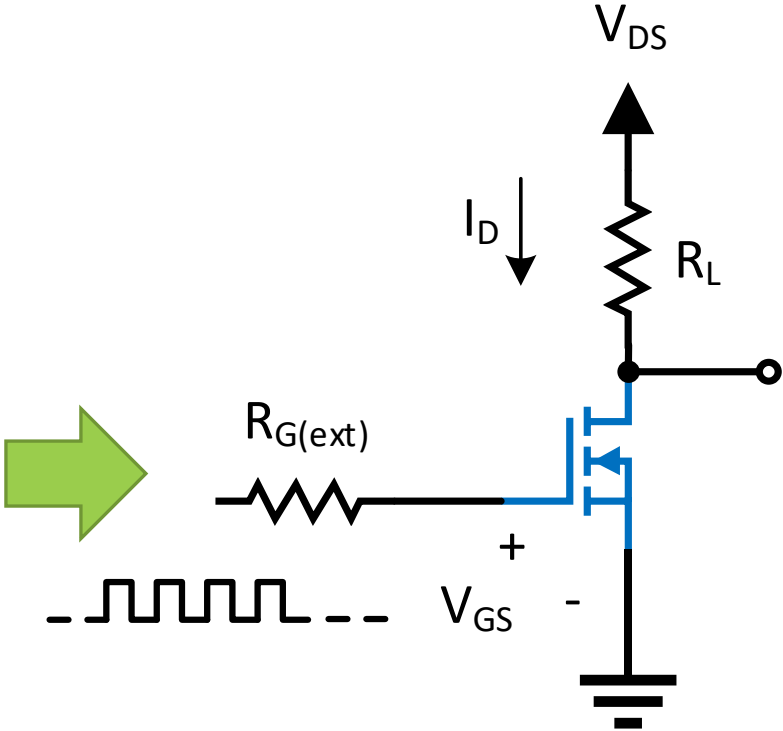
$$C_{iss} = C_{gs} + C_{gd}$$
$$C_{oss} = C_{ds} + C_{gd}$$
$$C_{rss} = C_{gd}$$

$t_{d(on)}$: the time from when the gate-source voltage rises over 10% of V_{GS} until the drain-source voltage reaches 90% of V_{DS}

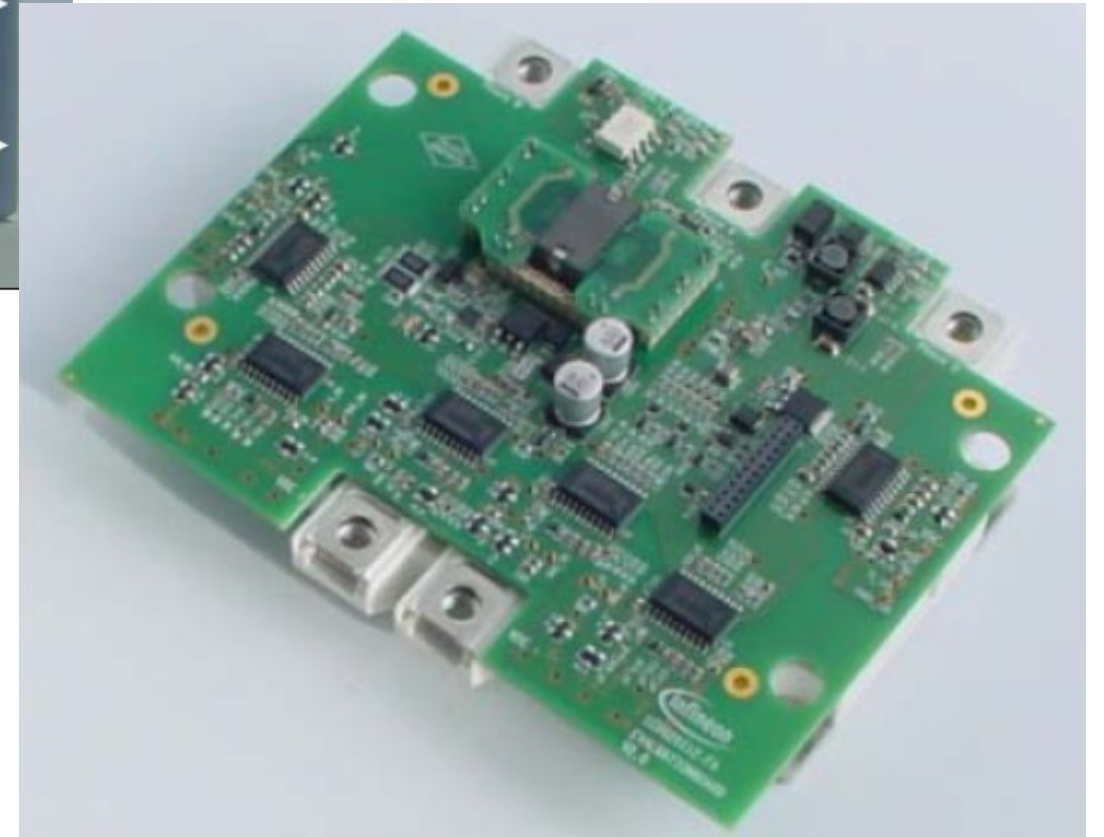
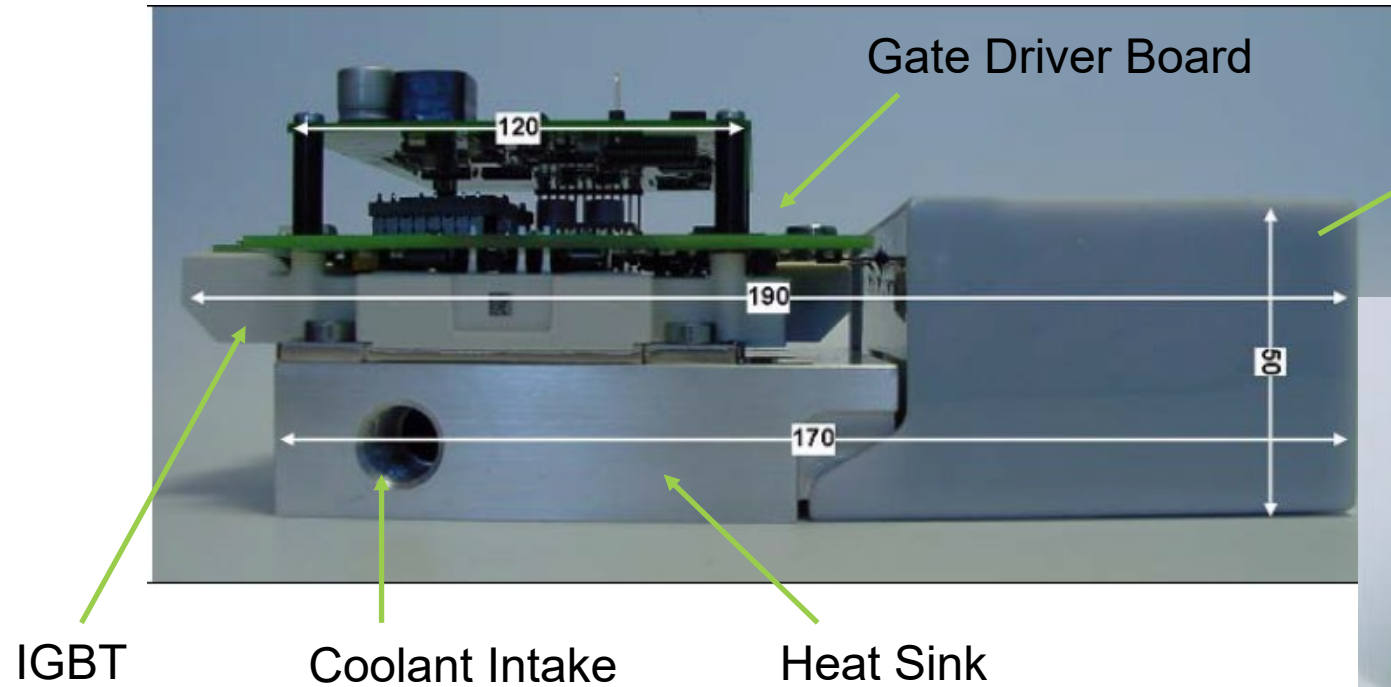
$t_{d(off)}$: the time from when the gate-source voltage drops below 90% of V_{GS} until the drain-source voltage reaches 10% of V_{DS}

t_r : the time taken for the drain-source voltage to fall from 90% to 10% of V_{DS}

t_f : the time taken for the drain-source voltage to rise from 10% to 90% of V_{DS}



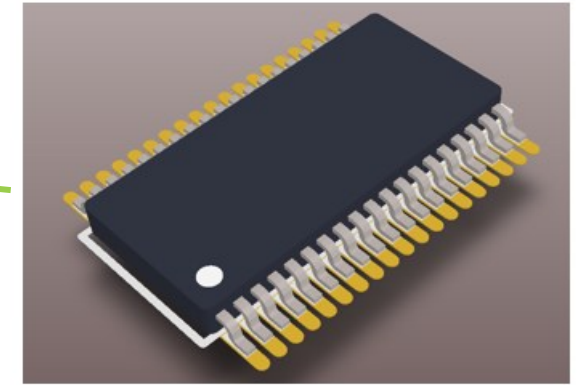
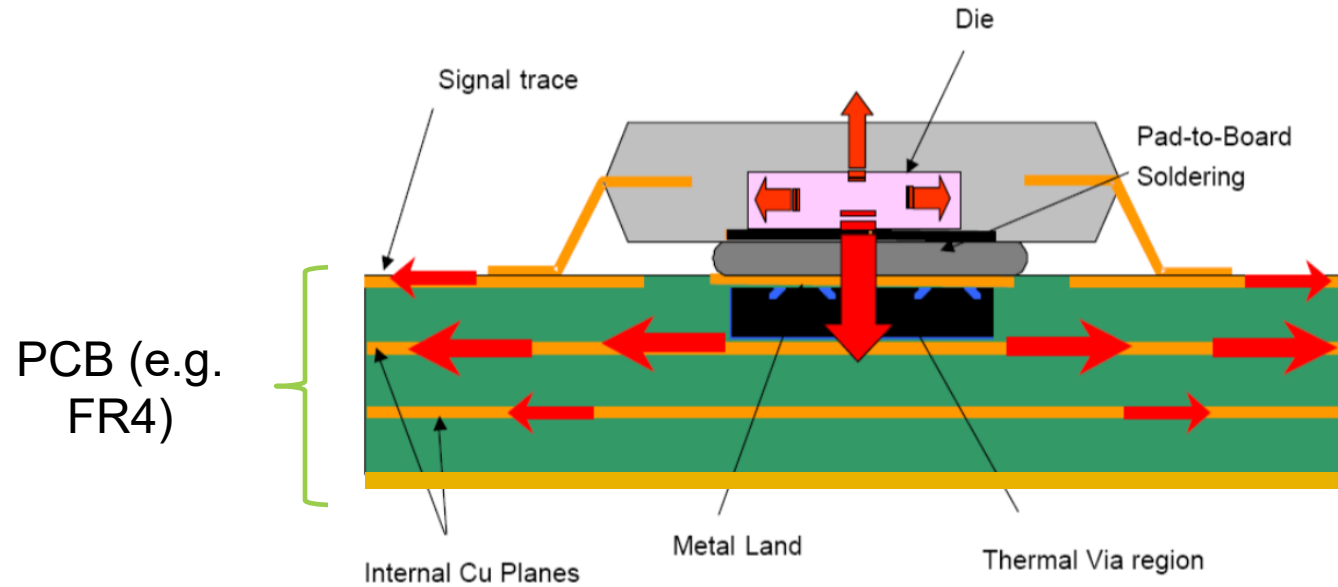
THERMAL CALCULATION



Infineon Hybrid-Kit Evaluation Driver Board

- Controller (microcontroller)
- Gate Driver
- IGBT Module

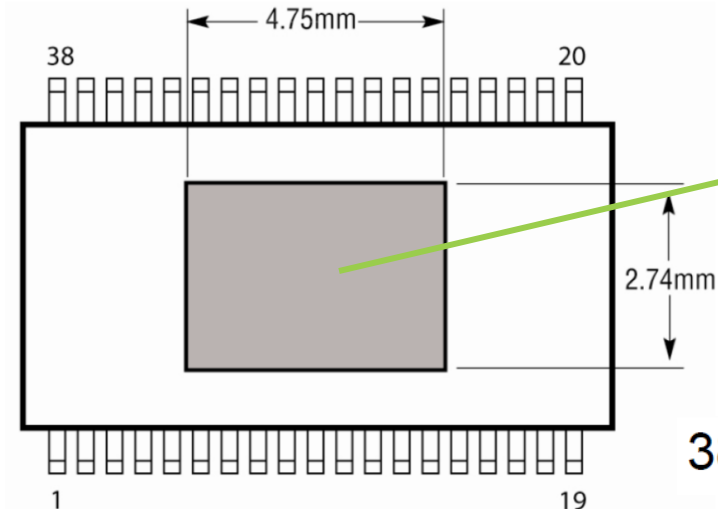
THERMAL CALCULATION



3D View of Package

Cross-section of exposed pad pkg with heat transfer

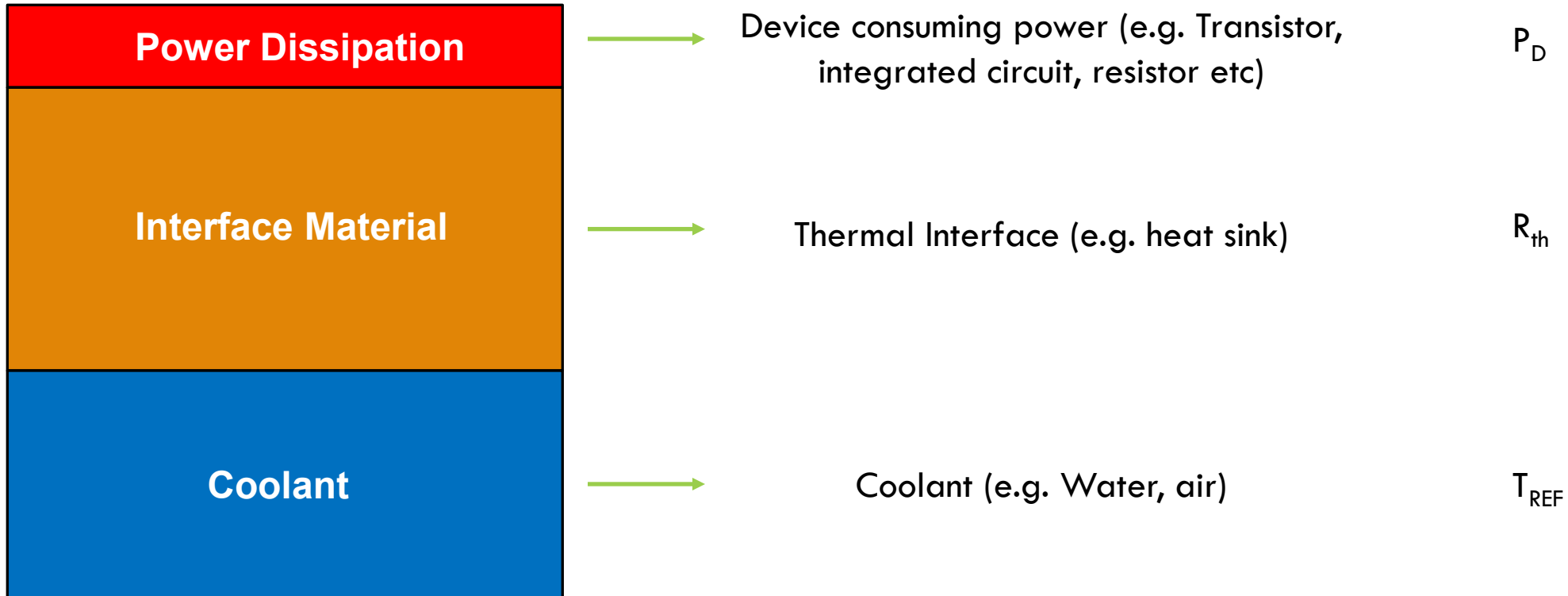
Bottom View of Package



Exposed Pad

38L TSSOP exposed pad pkg bottom view

THERMAL CALCULATION



P_D : Power Dissipation: If the current flowing through a $10\ \Omega$ resistor is 1 A, then P_D is 10 W $\rightarrow P_D = I^2.R$

R_{th} : Thermal Resistance (thermal impedance does not change over time)

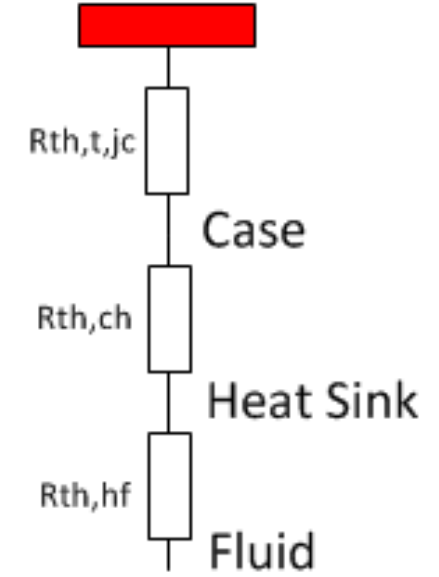
T_{REF} : Coolant temperature

If $P_D = 10\text{ W}$, $T_{REF} = 50\text{ }^\circ\text{C}$ and $R_{th} = 2\text{ }^\circ\text{C/W}$, then $T_{transistor} = T_{REF} + P_D.R_{th} = 70\text{ }^\circ\text{C}$

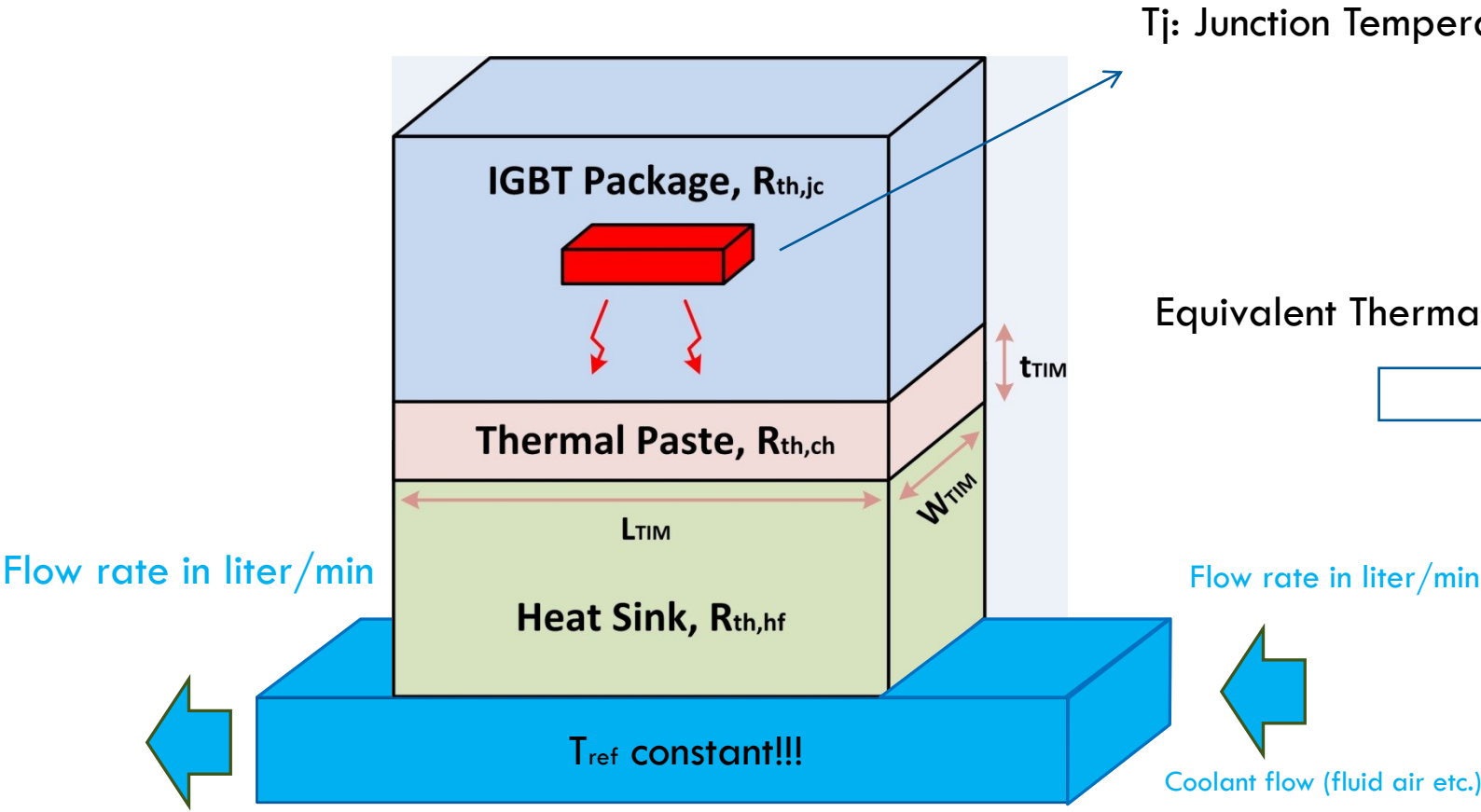
THERMAL CALCULATION

T_j : Junction Temperature

Semiconductor Device (e.g. IGBT)



Equivalent Thermal Circuit



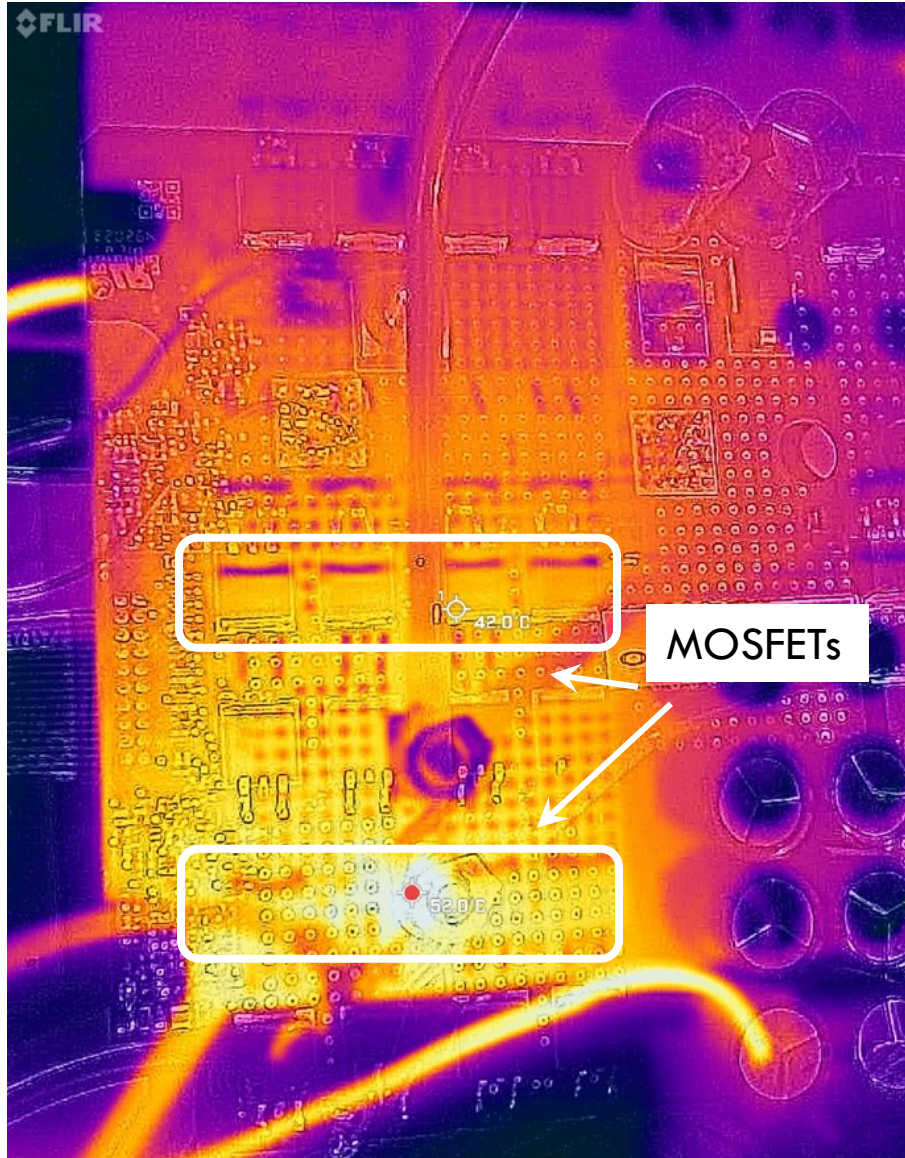
$$T_j = P_{diss}R_{th,jc} + P_{diss}R_{th,ch} + P_{diss}R_{th,hf} + T_{ref}$$

R_{th} is a thermal resistor (K/W or °C/W)

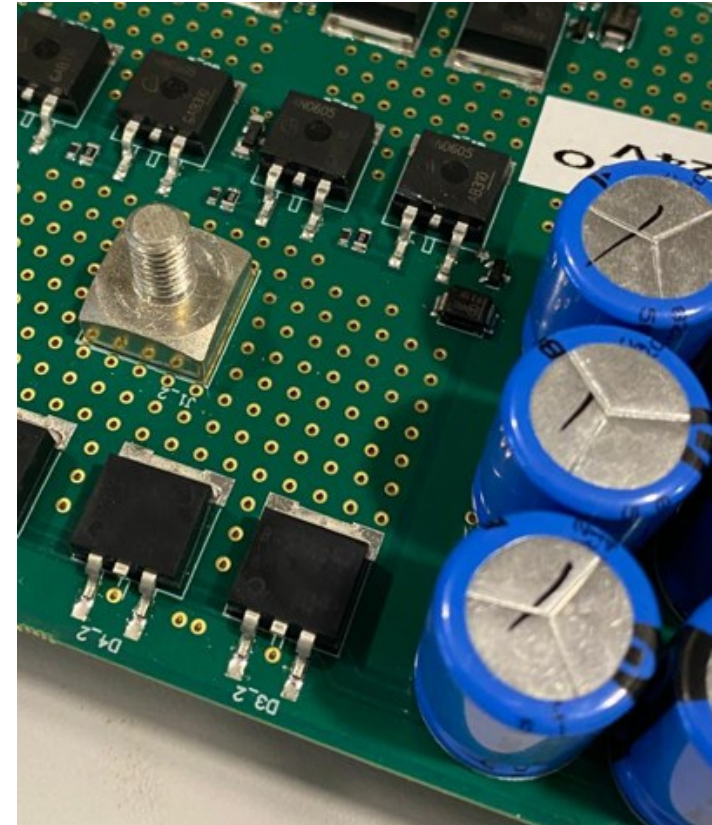
T_{ref} is the temperature of the coolant
(e.g. coolant fluid)

The propagation of heat vs. time (thermal capacitance) is not considered and will be neglected throughout this lecture!

MEASUREMENTS WITH A THERMAL CAMERA



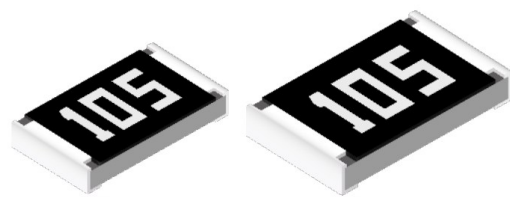
Each MOSFET has a Power Loss of 4.5 W



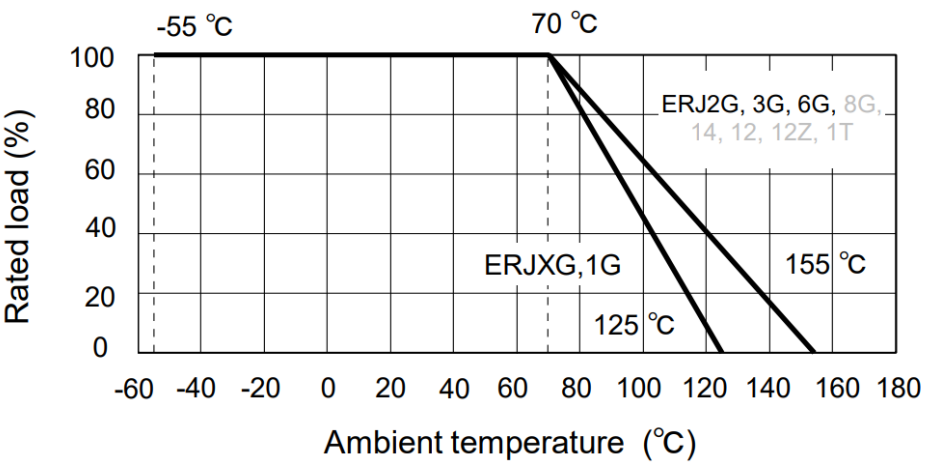
PASSIVE COMPONENTS

Resistors

SMD Resistors



Wire Wound Resistors



Aluminium Housed Wire Wound Resistors → can be mounted on a heat sink



Ceramic Wire Wound Power Resistors

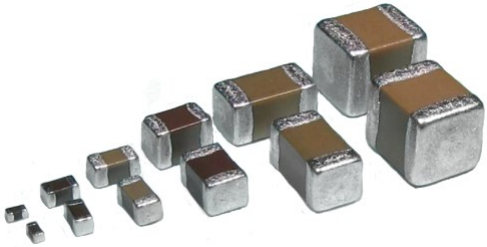
Part No. (inch size)	Rated power ^{*1} (70 °C) (W)	Resistance tolerance (%)	T.C.R. (×10 ⁻⁶ /K)	Category temperature range (°C)
ERJXG (01005)	0.031	±5	R<10Ω : -100 to +600 10Ω to 100Ω : ±300 100Ω≤R : ±200	-55 to +125

$$P_R = I^2 R$$

PASSIVE COMPONENTS

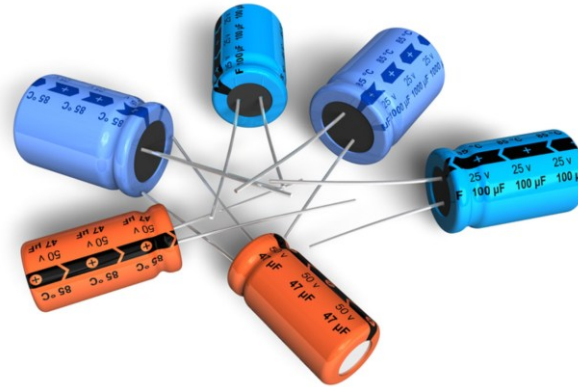
Capacitors

SMD Capacitors



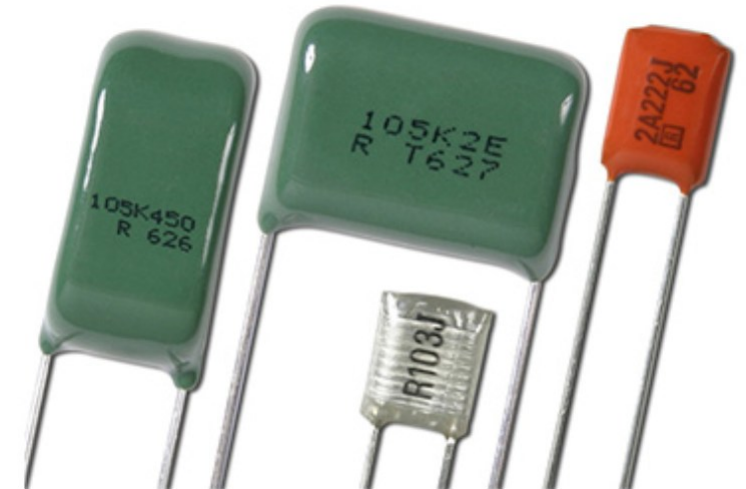
Ceramic capacitor: Small and stable, works well at high frequencies. Used for filtering and noise reduction in circuits.

Electrolytic Capacitors



Electrolytic capacitor: Stores a lot of charge but has polarity. Common in power supplies for smoothing voltage.

Film Capacitors



Film capacitor: Uses plastic film, very stable with low losses. Common in power electronics for DC-link, snubbers, and EMI filters.

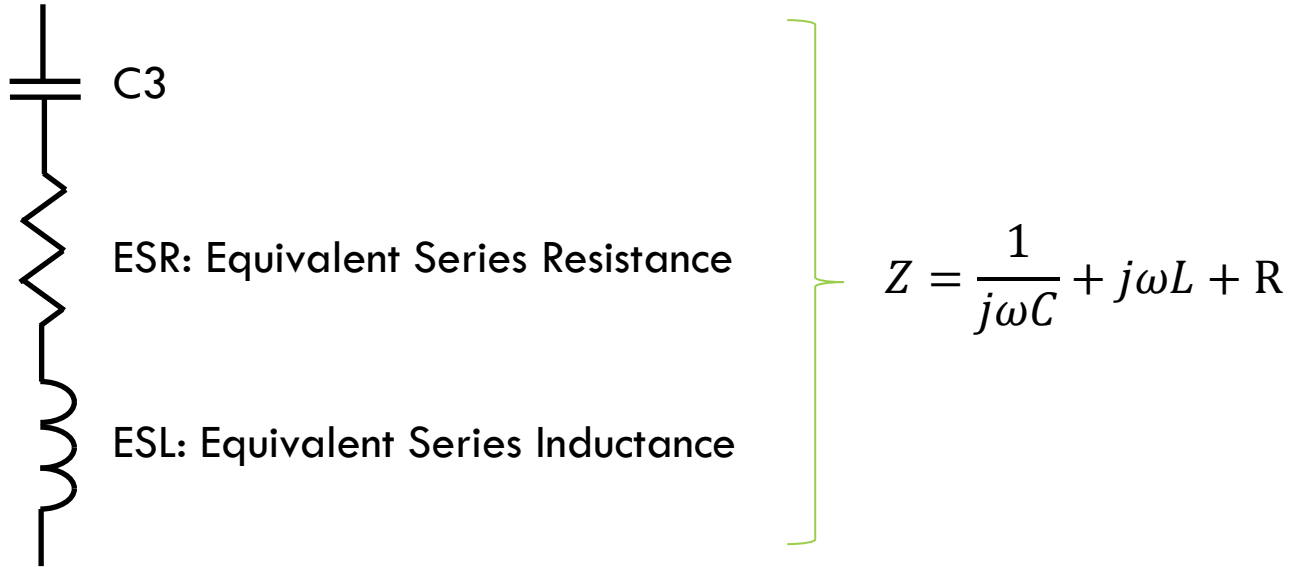
PASSIVE COMPONENTS

Capacitors

Feature	Ceramic Capacitor	Electrolytic Capacitor	Film Capacitor
Dielectric	Ceramic material	Electrolyte + oxide layer	Plastic film (PP, PET, etc.)
Capacitance Range	pF – μ F (low to medium)	μ F – thousands of μ F (very high)	nF – tens of μ F (low to medium-high)
Polarity	Non-polarized	Polarized	Non-polarized
Stability	Very stable, low losses	Moderate, drifts with age/temp	Excellent, highly stable
ESR (Series R)	Very low	High	Low to medium
ESL (Series L)	Very low (compact size)	High (wound structure)	Low to medium
High-Freq Behavior	Excellent, MHz range capable	Poor, only low-freq filtering	Good, up to 100s kHz–MHz
Lifetime	Very long	Limited (electrolyte dries)	Long, durable
Applications	Decoupling, RF, filters	Power supply smoothing, storage	DC-link, snubbers, EMI, audio

PASSIVE COMPONENTS

Capacitors



ESR causes power loss if an AC current flows through the capacitor. Therefore, when selecting the capacitor, the maximum ripple current becomes very important!

$$\frac{1}{j\omega C} = -j\omega C \quad \Rightarrow \quad \text{At a specific frequency called resonance frequency: } -j\omega_R C = j\omega_R L$$

At resonance frequency, C3 is no more a capacitor!

PASSIVE COMPONENTS

Capacitors



C3

ESR: Equivalent Series Resistance

ESL: Equivalent Series Inductance



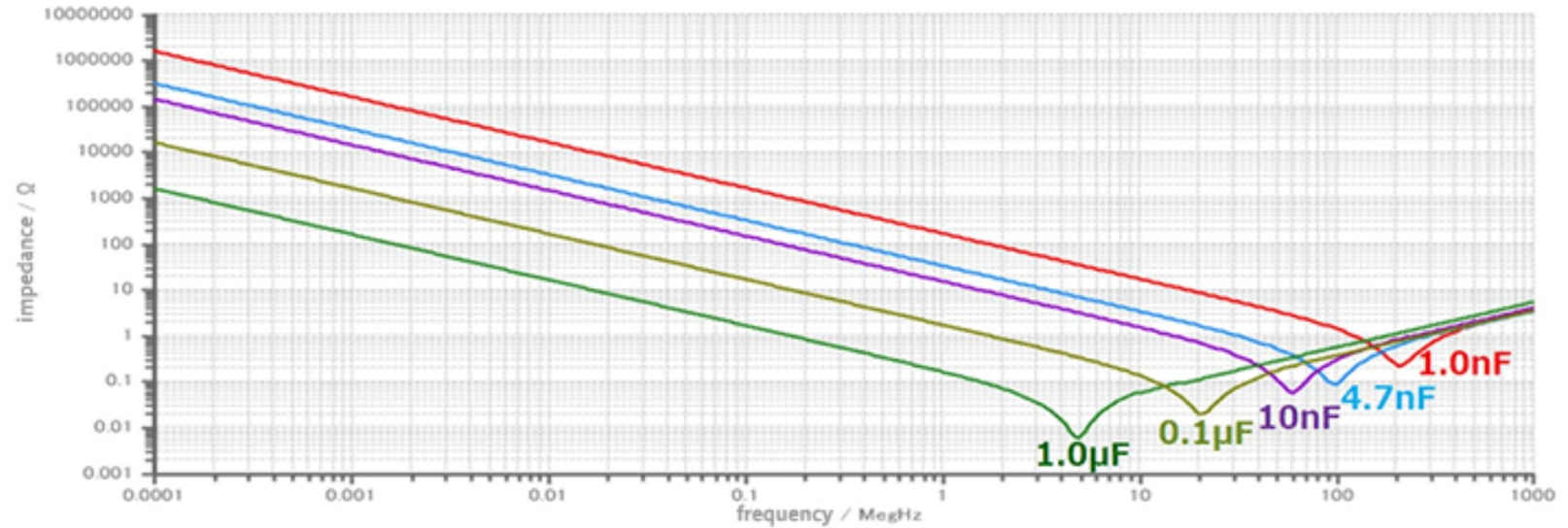
$$Z = \frac{1}{j\omega C} + j\omega L + R$$

$$\frac{1}{j\omega C} = -j\omega C$$



At a specific frequency called resonance frequency: $-j\omega_R C = j\omega_R L$

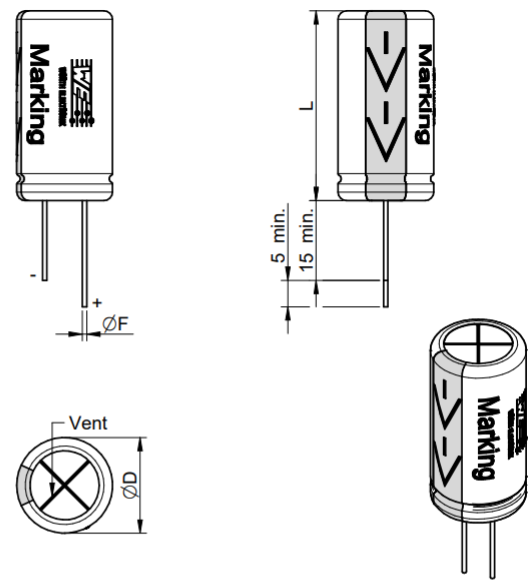
At resonance frequency, C3 is no more a capacitor!



PASSIVE COMPONENTS

Capacitors - Datasheet

Example: Eletrolytic Capacitor



Properties		Value	Unit	Tol.
Diameter	Ø D	18	mm	±0.5
Length	L	21	mm	±1.5
Pitch		7.5	mm	±0.5
Pin Diameter	Ø F	0.8	mm	±0.02
Pitch	p	7.5	mm	±0.5
Hole Diameter	Ø f	1.1	mm	±0.02

General Information:

Aluminum Electrolytic Capacitors	
Storage Conditions	5-35 °C, < 75% RH
Operating Temperature	-25 °C up to +105 °C
Endurance	10000 h @ / 450 V (DC) / max. Iripple

Electrical Properties:

Properties	Test conditions		Value	Unit	Tol.
Capacitance	0.25 V/ 120 Hz/ +20 °C	C	33	µF	±20%
Rated Voltage		U _R	450	V (DC)	max.
Ripple Current	100 kHz @ 105 °C	I _{RIPPLE}	883	mA	max.

$$P_{\text{loss}} = I_{\text{rms}}^2 \cdot R_{\text{ESR}}$$

Ripple current in a capacitor is the AC current component that flows through the capacitor when it is used in a circuit with a varying or pulsating voltage.

If the ripple current exceeds the rated value:

- The capacitor temperature rises,
- Internal losses increase,
- The expected lifetime is significantly reduced.

Therefore, the ripple current rating represents a thermal limit and is a critical parameter in power electronics design.

PASSIVE COMPONENTS

X Capacitors

A class X capacitor is designed to go across a mains input (from line to neutral or from phase to phase). It is assumed that there is some form of current limiting in the supply, such as a fuse or over-current trip, so if the capacitor fails short circuit the protection device will open. Therefore, if X class capacitors fail short circuit, as long as the capacitor does not catch fire or ignite any other adjacent components, it will then fail safe. Using open-mode capacitors significantly lowers the risk of short-circuit failures.

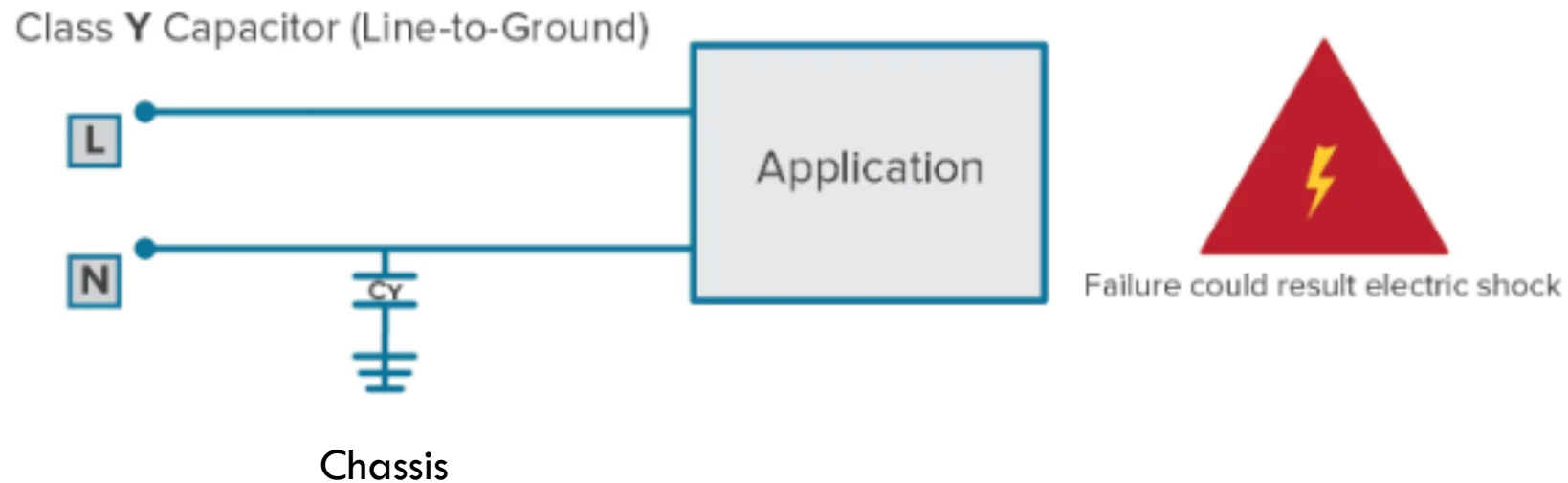
Class X Capacitor (Line-to-Line)



PASSIVE COMPONENTS

Y Capacitors

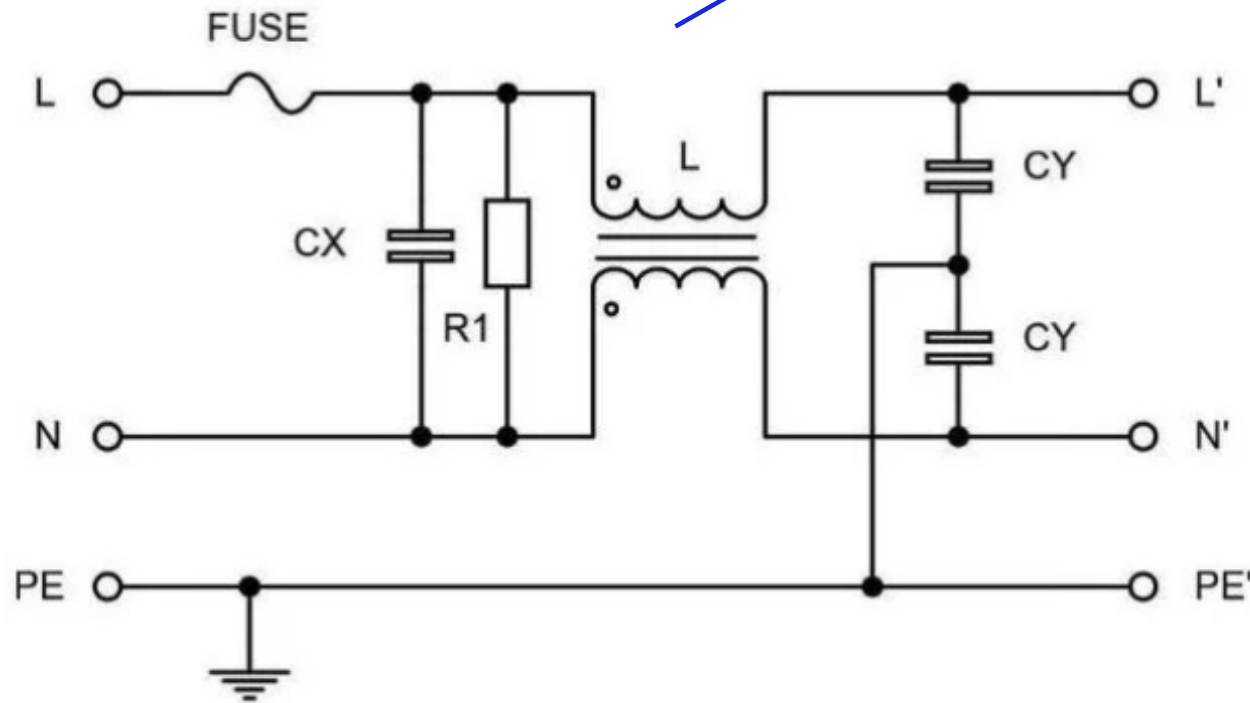
A class Y capacitor is designed to go from line to ground line to a zero potential (fuse will not work) or across the isolation barrier from primary to the secondary side. A short circuit would lead to hazardous voltages appearing on the zero potential or secondary side connections, so they are designed to fail open circuit. For many applications, a double fault must not cause an unacceptable hazard, so for medical and household applications, two Y-class capacitors in series are specified across the isolation barrier or from line to ground/zero potential.



PASSIVE COMPONENTS

X and Y Capacitors → why do we need them?

Common-Mode Choke: A common-mode choke is an inductor with two windings on the same core that blocks high-frequency noise currents flowing in the same direction on multiple lines. It allows the desired differential signal to pass while suppressing common-mode EMI in power and signal circuits.



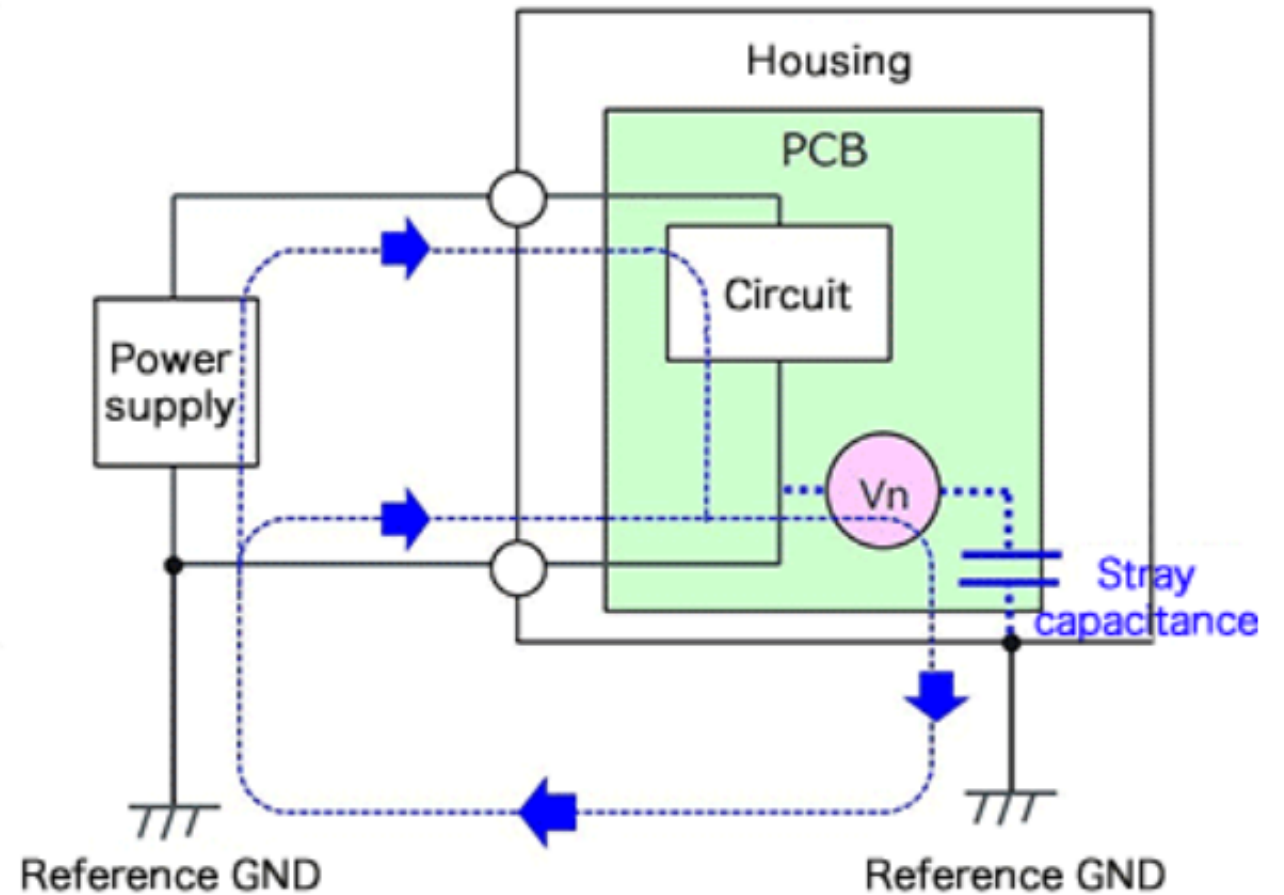
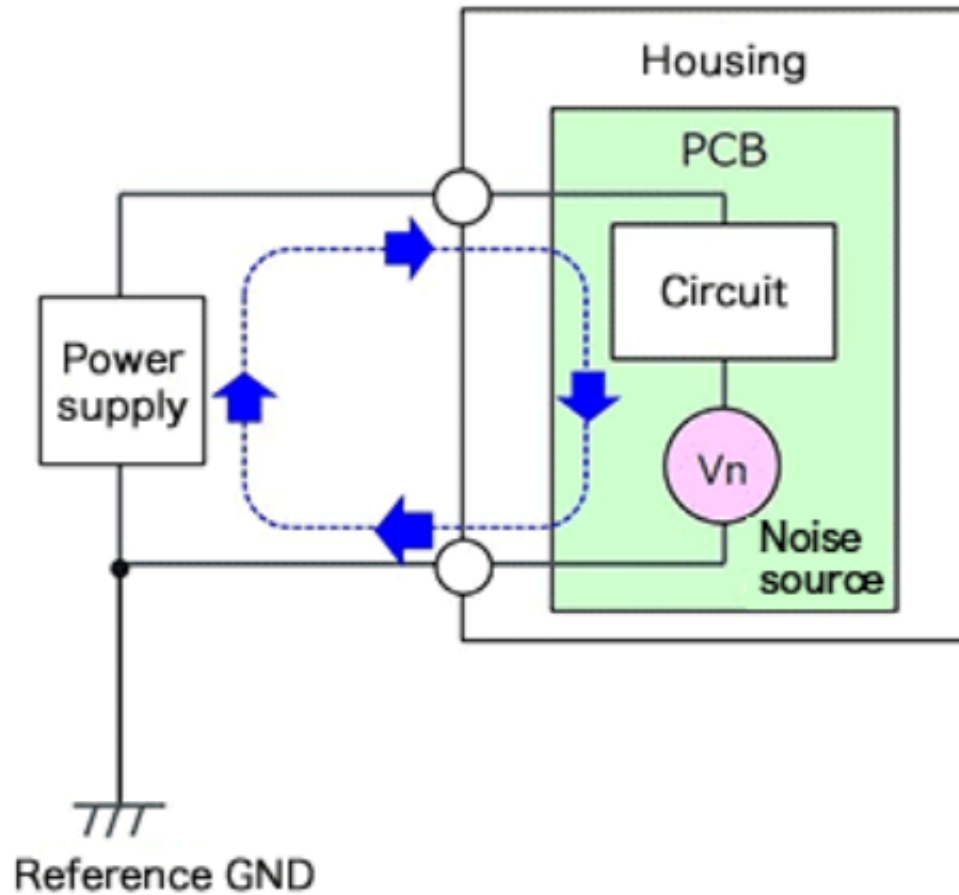
LOAD:

Switched-Mode Power Supply

Common-Mode and Differential Noise at next slide!

PASSIVE COMPONENTS

Common-Mode and Differential Noise



PASSIVE COMPONENTS

Inductors

SMD Inductors



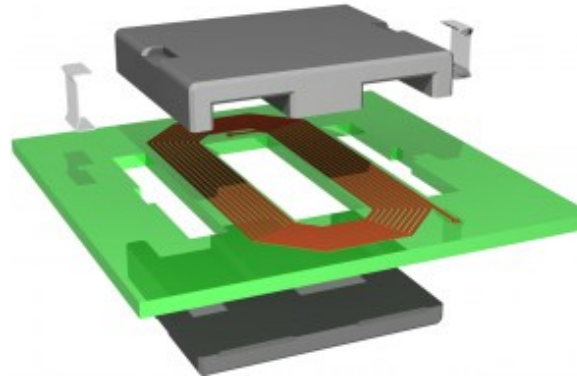
Packaged in small surface-mount form for compact PCB design. Used in RF circuits and noise filtering; ferrite beads suppress high-frequency EMI.

Wound Inductors



Made by winding copper wire around a ferrite or iron powder core. It offers high inductance and current handling, making it common in power supply circuits.

Planar Inductors



Formed directly by PCB copper traces in a spiral pattern. Compact and cost-effective, mostly used in RF matching networks and integrated DC-DC converters.

Torodial Inductors



Wound around a torus-shaped ferrite or iron core. Provides high efficiency and low EMI, widely used in power electronics for filtering.

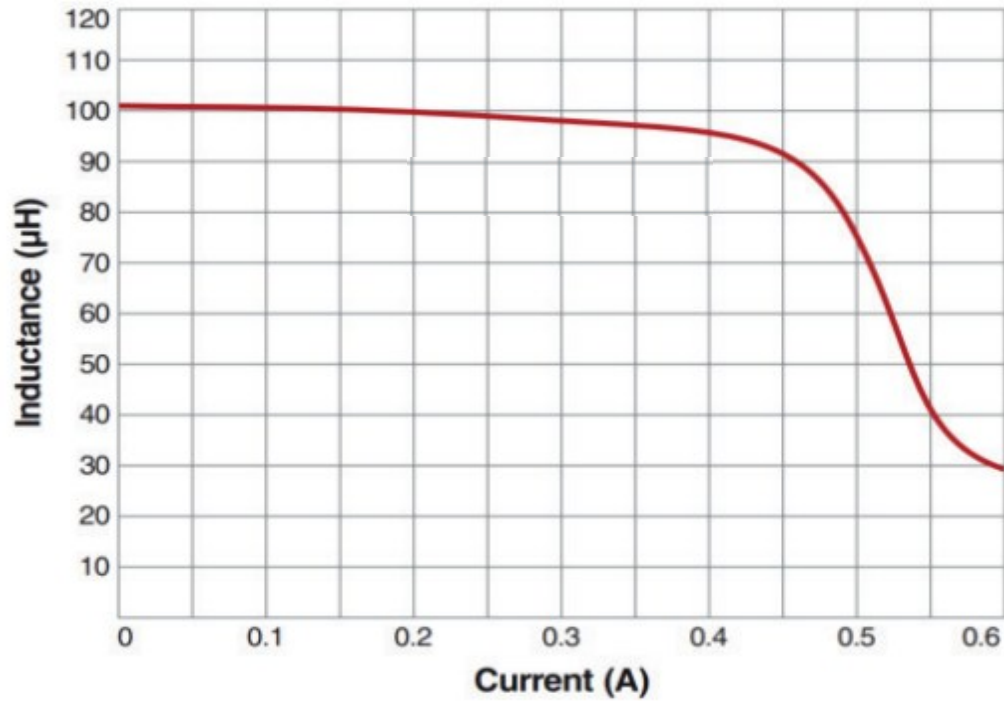
PASSIVE COMPONENTS

Inductors

Inductor Type	Inductance Range	Max Current	ESR (Series R)	ESL (Series L)	Typical Frequency Range
Wire-Wound Inductor	μH – mH	High (up to 10–50 A)	Low ($\text{m}\Omega$ range)	Medium	kHz – few MHz (power apps)
SMD Chip Inductor	nH – μH	Low–Medium (0.1–5 A)	Very low	Very low	MHz – GHz (RF, decoupling)
Planar (PCB Spiral)	nH – low μH	Low–Medium ($\leq$ 2–3 A)	Higher (Ω range)	Medium–High	MHz (RF, antennas, matching)
Toroidal Inductor	μH – tens of mH	Very High (10–100 A)	Very low ($\text{m}\Omega$ range)	Low–Medium	kHz – hundreds of kHz

PASSIVE COMPONENTS

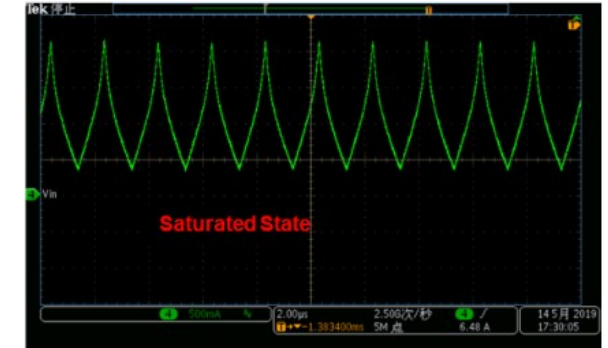
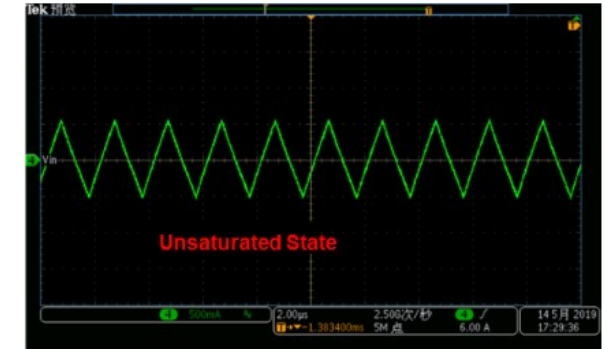
Inductor Saturation



One effect of current through an inductor is core saturation.

Frequently DC-DC converters have current wave shapes with a dc component. The dc current through an inductor biases the core and can cause it to become saturated with magnetic flux. The designer needs to understand that when this occurs the inductance drops and the component no longer functions as an inductor.

Current Probe Measuring Current Flowing Through an Inductor Soldered on a PCB



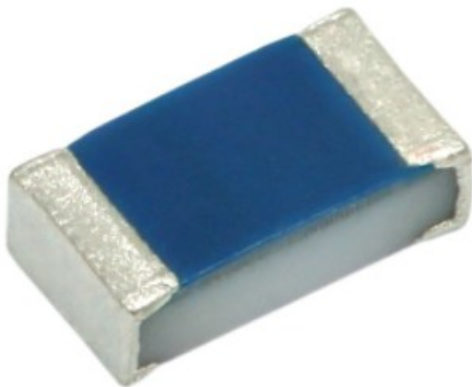
Current flowing through the inductor

TEMPERATURE SENSORS

A **thermistor** is a combination of two words: Thermal and Resistor, which literally makes it a temperature-dependent resistor.



The NTC is the most commonly used thermistor, especially the 10K Ω NTC thermistor. It is also popular due to its reliability and quick response. NTC Resistance decreases as temperature increases.



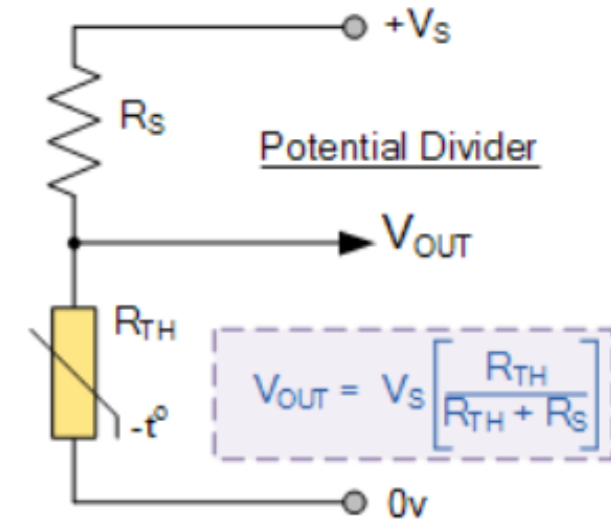
The PTC has a totally opposite usage than an NTC, though they are not as commonly used, they are usually used for self-regulating heating elements/self-resetting purposes (fuse). Some of the characteristics it possesses are:

- Resistance increases as the temperature increases.
- Acts as a throttle in a circuit.
- Exhibits a sudden increase in resistance above a defined temperature.

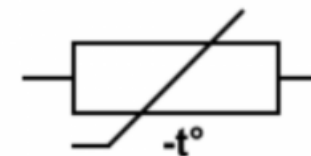
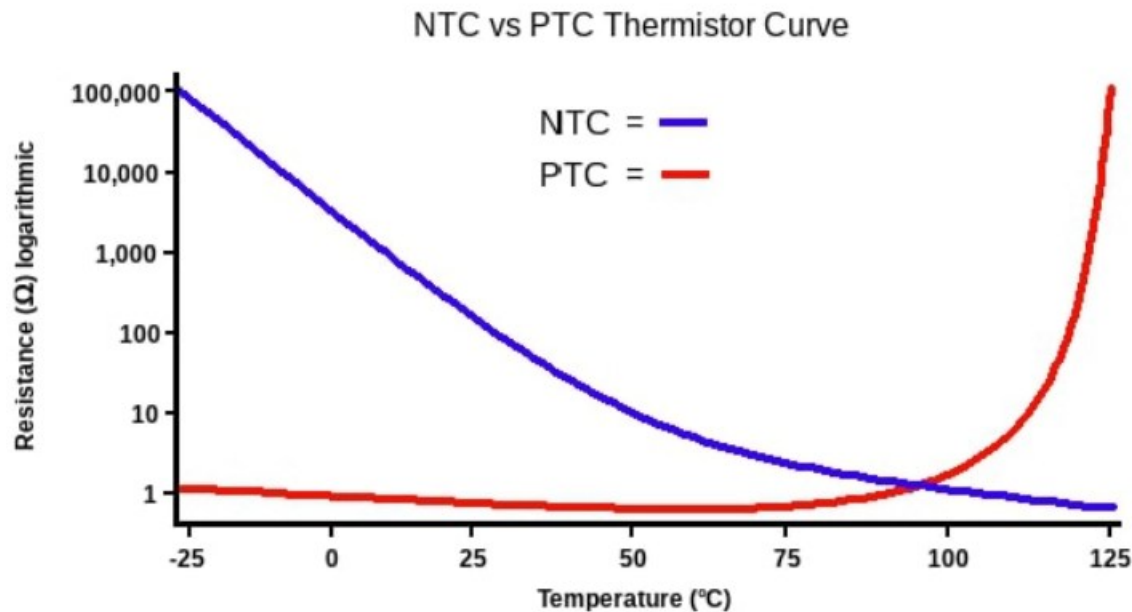
TEMPERATURE SENSORS

NTC vs PTC Thermistors

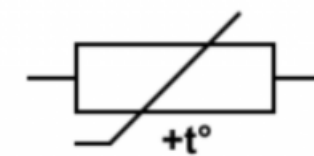
Thermistors	NTC	PTC
Temperature coefficient	Negative (-ve)	Positive (+ve)
Metal Oxides	Nickel, iron, manganese, titanium, cobalt	Strontium titanate, barium-, lead-
Temperature Range	-55°C to 200°C	60°C to 120°C
Applications	Temperature sensing and control, flow measurement etc.	Over-current protection, self-regulating heater etc.



If you use a standard $10\text{k}\Omega$ thermistor with a series resistor of $10\text{k}\Omega$, the output voltage at the base temperature of 25°C will be half the supply voltage as $10\text{k}\Omega / (10\text{k}\Omega + 10\text{k}\Omega) = 0.5V$ if $V_s = 1V$

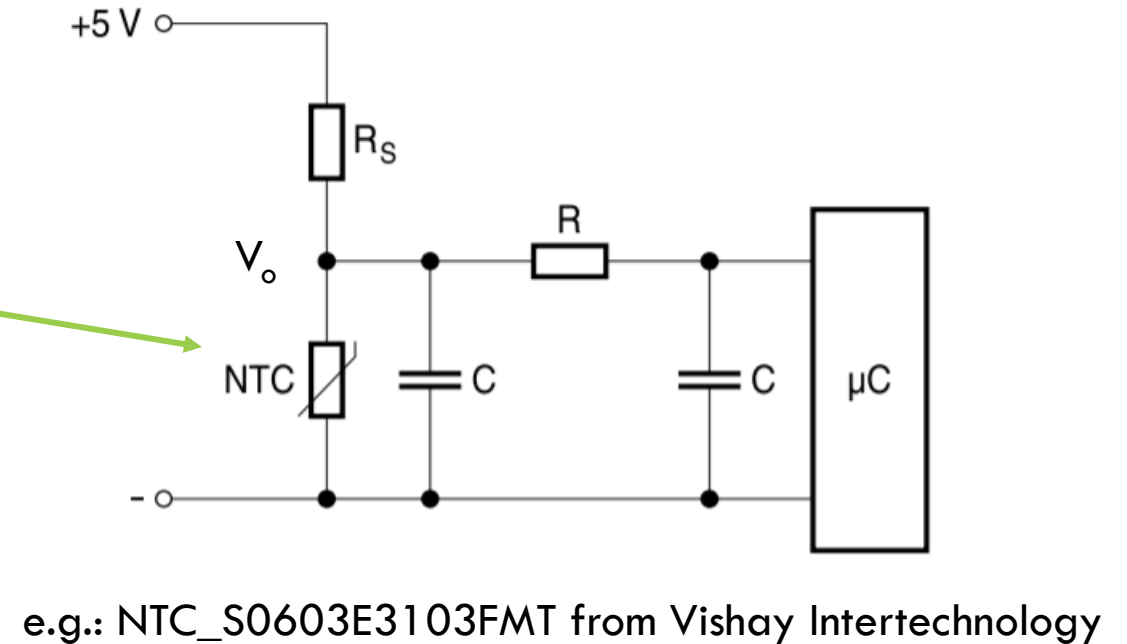
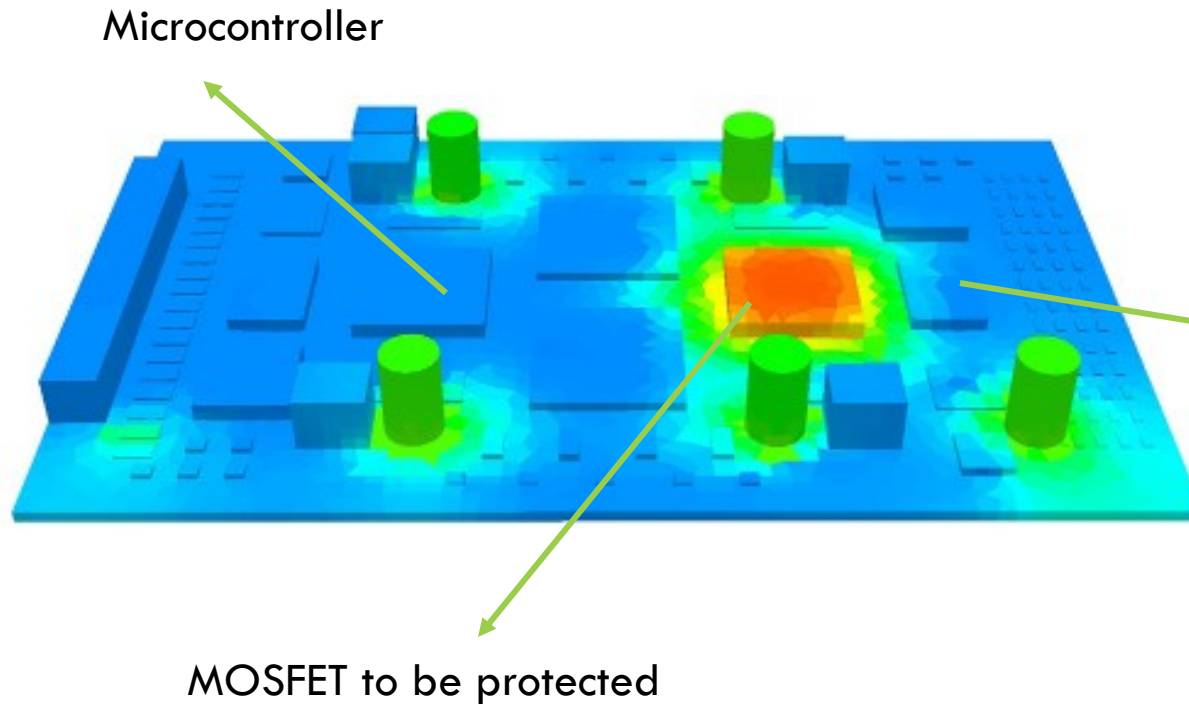


NTC thermistor symbol



PTC thermistor symbol

TEMPERATURE SENSORS – APPLICATION EXAMPLE



NTC temperature sensor placed next to the MOSFET for temperature measurement. The microcontroller shall turn off the MOSFET as soon as the MOSFET temperature exceeds 150°C

TEMPERATURE SENSORS – APPLICATION EXAMPLE

Temperature T	R _T	R _{pullup}	V _{supply}	V _o
[°C]	[Ohm]	[Ohm]	[V]	[V]
-55.0	636177.186	10000	5.00	4.92
-50.0	456420.386	10000	5.00	4.89
-45.0	331435.765	10000	5.00	4.85
-40.0	243448.302	10000	5.00	4.80
-35.0	180771.935	10000	5.00	4.74
-30.0	135622.815	10000	5.00	4.66
-25.0	102751.480	10000	5.00	4.56
-20.0	78575.541	10000	5.00	4.44
-15.0	60622.555	10000	5.00	4.29
-10.0	47167.676	10000	5.00	4.13
-5.0	36995.188	10000	5.00	3.94
0.0	29239.781	10000	5.00	3.73
5.0	23279.759	10000	5.00	3.50
10.0	18664.427	10000	5.00	3.26
15.0	15064.246	10000	5.00	3.01
20.0	12236.276	10000	5.00	2.75
25.0	10000.000	10000	5.00	2.50
30.0	8220.272	10000	5.00	2.26
35.0	6795.174	10000	5.00	2.02
40.0	5647.315	10000	5.00	1.80
45.0	4717.529	10000	5.00	1.60
50.0	3960.296	10000	5.00	1.42
55.0	3340.375	10000	5.00	1.25
60.0	2830.323	10000	5.00	1.10
65.0	2408.644	10000	5.00	0.97
70.0	2058.410	10000	5.00	0.85
75.0	1766.218	10000	5.00	0.75
80.0	1521.400	10000	5.00	0.66
85.0	1315.427	10000	5.00	0.58
90.0	1141.443	10000	5.00	0.51
95.0	993.914	10000	5.00	0.45
100.0	868.354	10000	5.00	0.40
105.0	761.107	10000	5.00	0.35
110.0	669.186	10000	5.00	0.31
115.0	590.137	10000	5.00	0.28
120.0	521.939	10000	5.00	0.25
125.0	462.917	10000	5.00	0.22
130.0	411.683	10000	5.00	0.20
135.0	367.079	10000	5.00	0.18
140.0	328.138	10000	5.00	0.16
145.0	294.048	10000	5.00	0.14
150.0	264.124	10000	5.00	0.13

If $V_o = 0.13\text{ V}$, this corresponds to a MOSFET temperature of 150°C . The MOSFET is then switched off.

The ADC of the microcontroller continuously measures V_o to monitor the MOSFET temperature

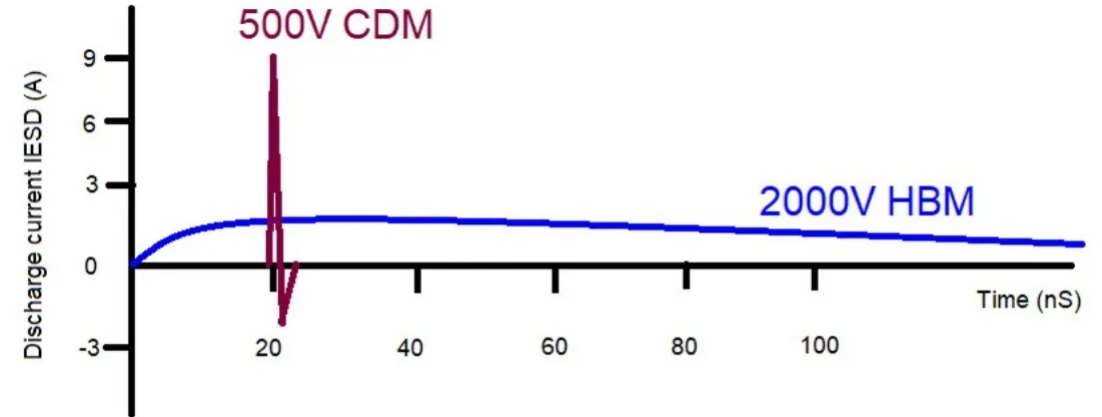
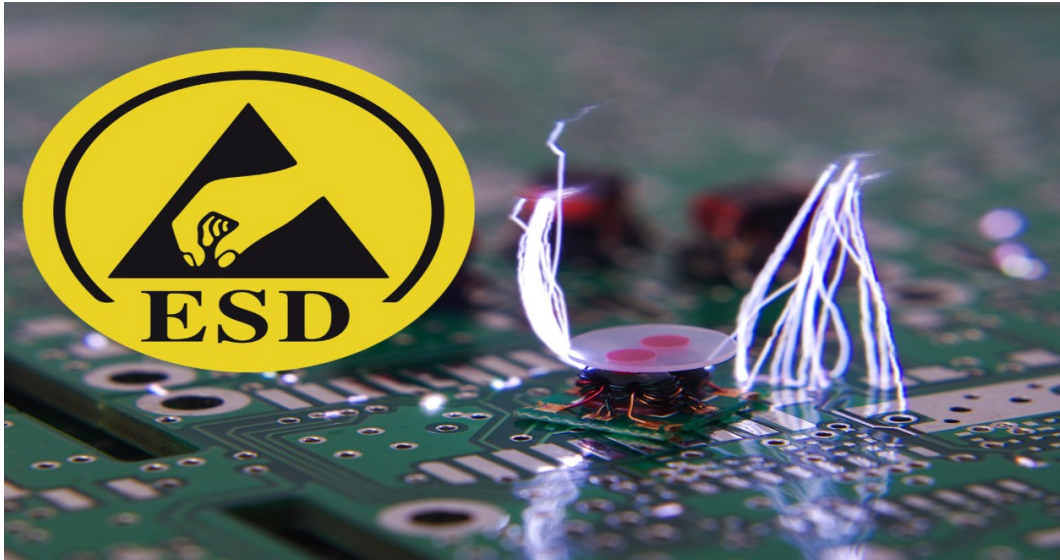
Note: At 25°C , the NTC resistance is $10\text{k}\Omega$ which corresponds to a V_o voltage of 2.5 V .

Because NTC thermistors typically have relatively large thermal time constants, they may respond too slowly to detect overcurrent events and the resulting rapid temperature rise. As a result, they can be inadequate for fast fault scenarios.

Therefore, in circuits that require rapid protection, direct overcurrent detection methods are generally more appropriate than relying on NTC-based temperature sensing.

DIODE CIRCUITS

ESD Protection



HBM: Human Body Model

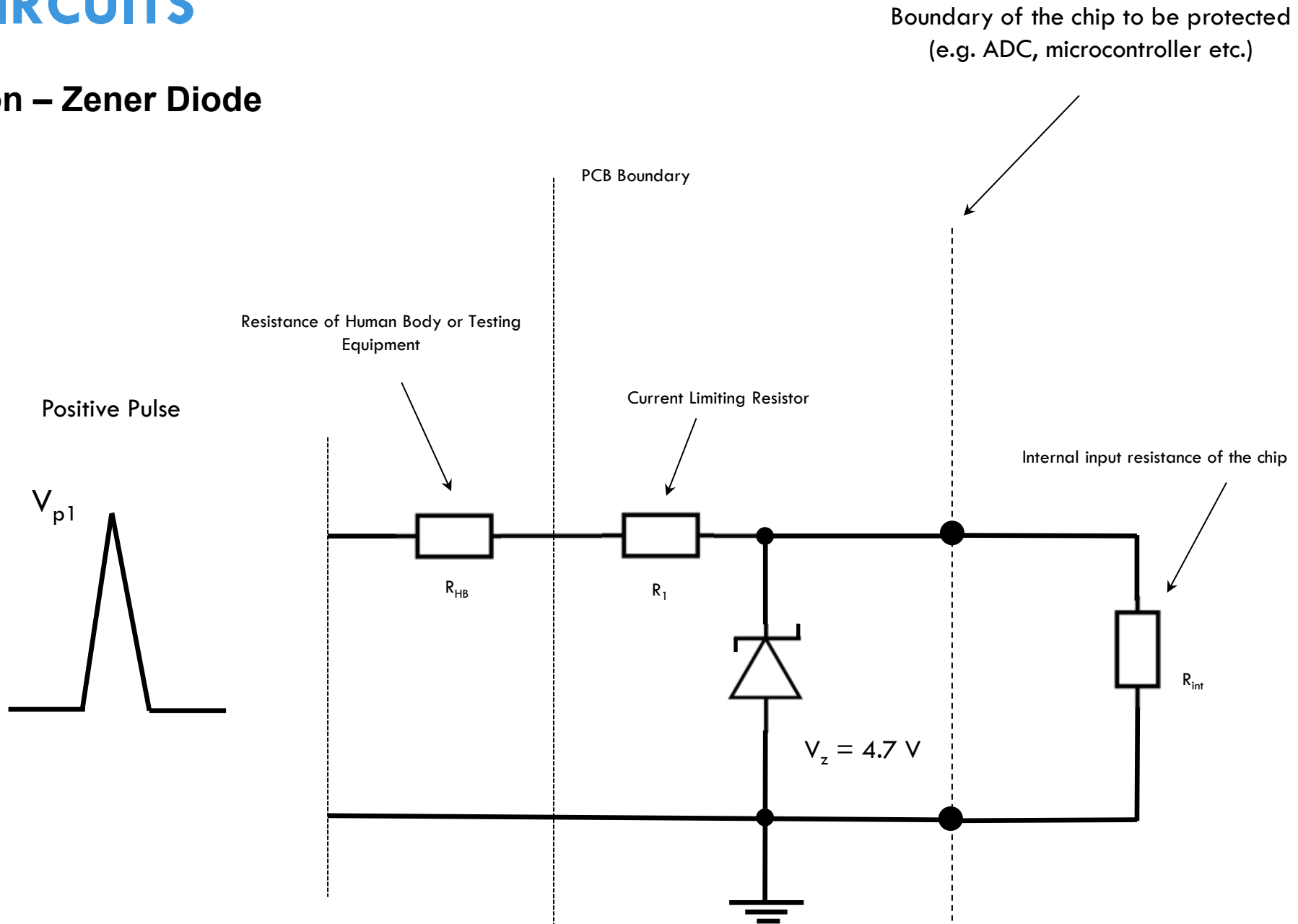
CDM: Charge Discharge Model

Electrostatic discharge (ESD) can unintentionally harm costly electronic equipment, leading either to immediate catastrophic failure or to premature breakdowns in the field. In electronics manufacturing—particularly in today’s semiconductor-driven era—even small static charges can severely impact product yield and lifespan.

ESD is the sudden release of accumulated charge between objects, allowing electricity to pass from one to the other. This discharge, often originating from statically charged humans or objects, is powerful enough to damage many semiconductor components, including MOSFETs, diodes, and CMOS ICs such as CPUs and graphics processors.

DIODE CIRCUITS

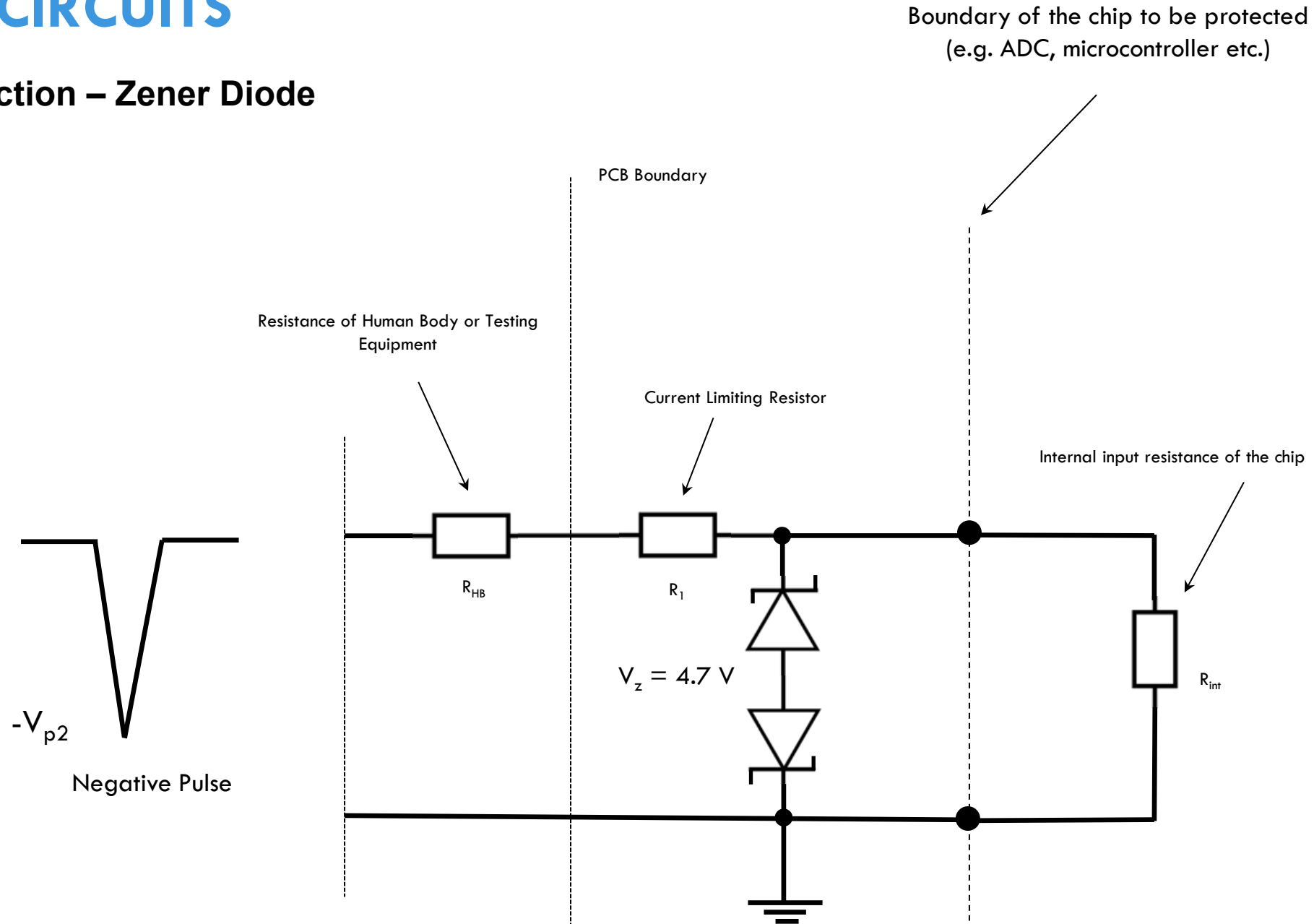
ESD Protection – Zener Diode



This protection method is suitable only for signal lines and not for supply lines due to the voltage drop on R_1 !

DIODE CIRCUITS

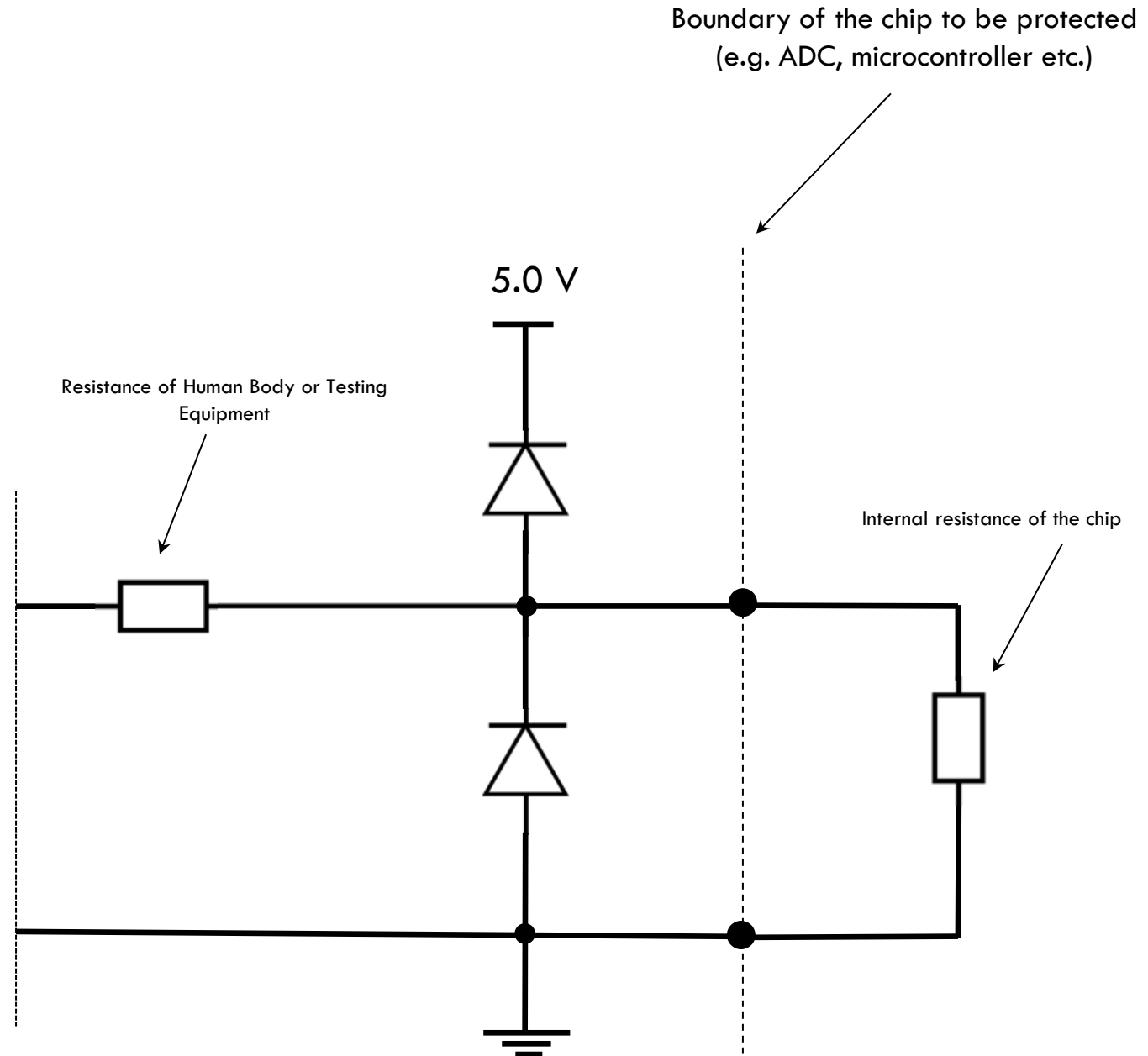
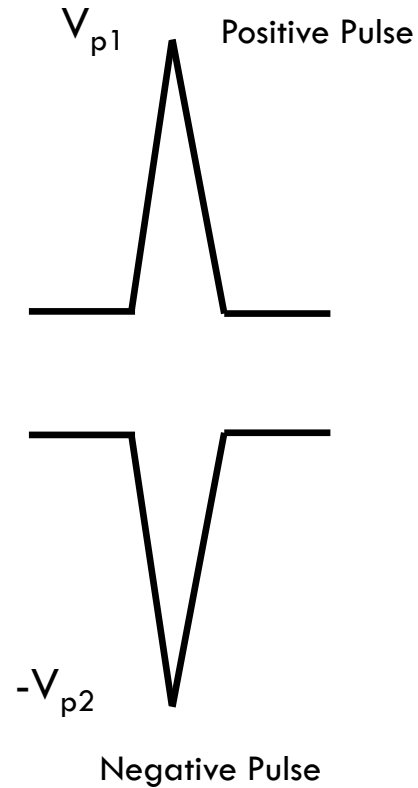
ESD Protection – Zener Diode



This protection method is suitable only for signal lines and not for supply lines due to the voltage drop on R_1 !

DIODE CIRCUITS

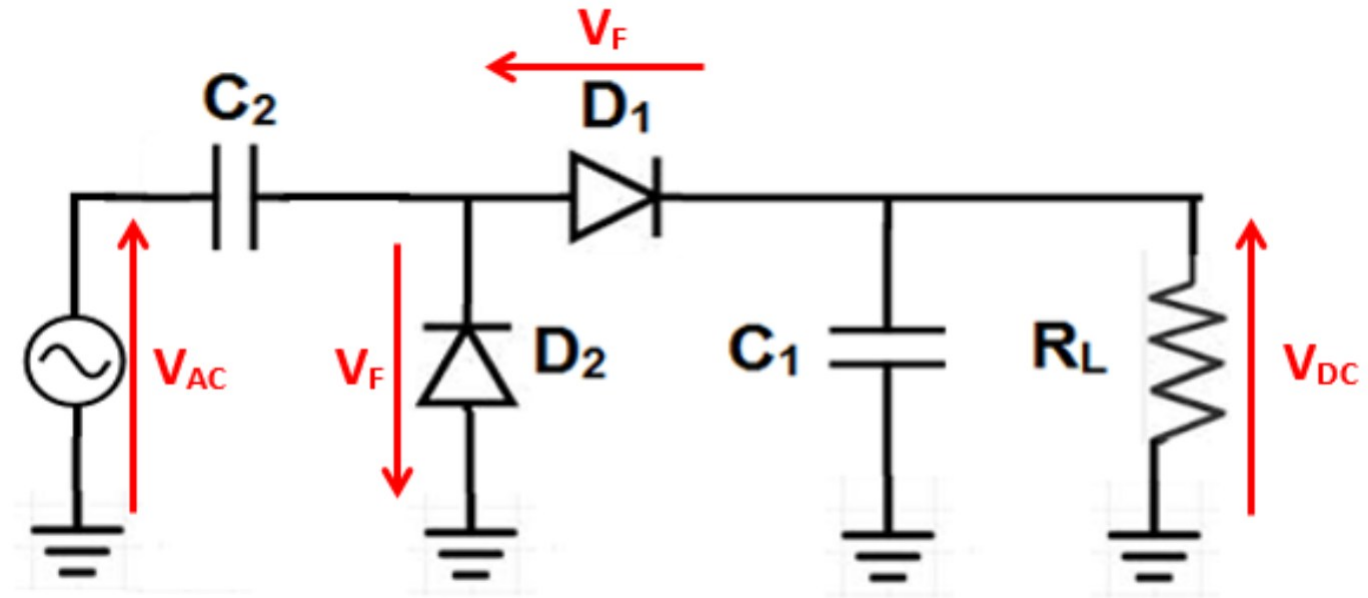
ESD Protection – Diode



DIODE CIRCUITS

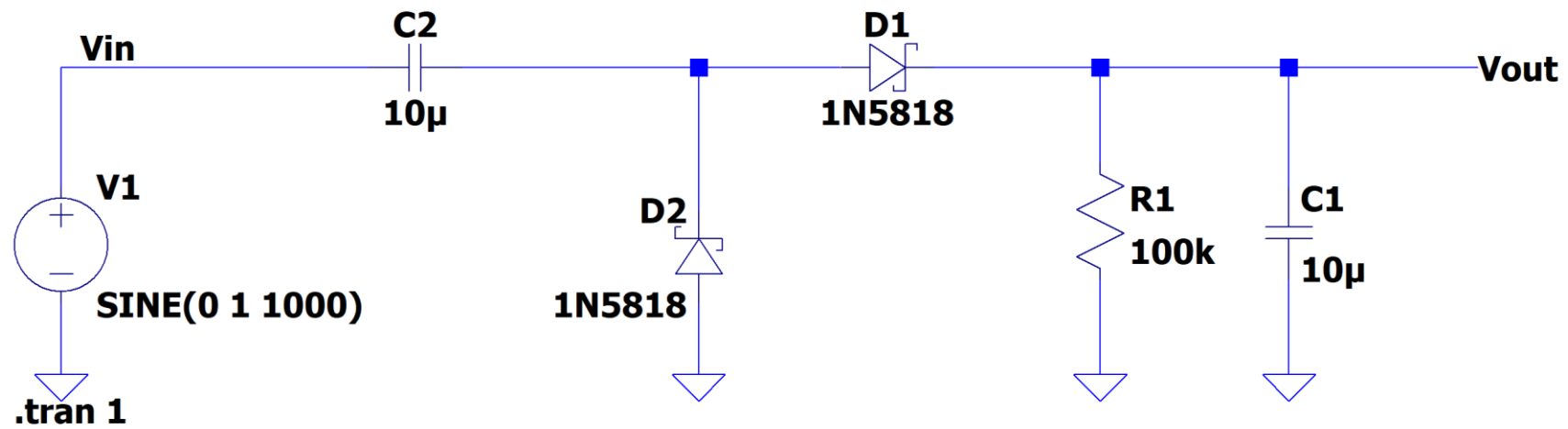
Diode Voltage Doubler

Analyze the operation of the circuit shown below, assuming the input signal is a sine wave.



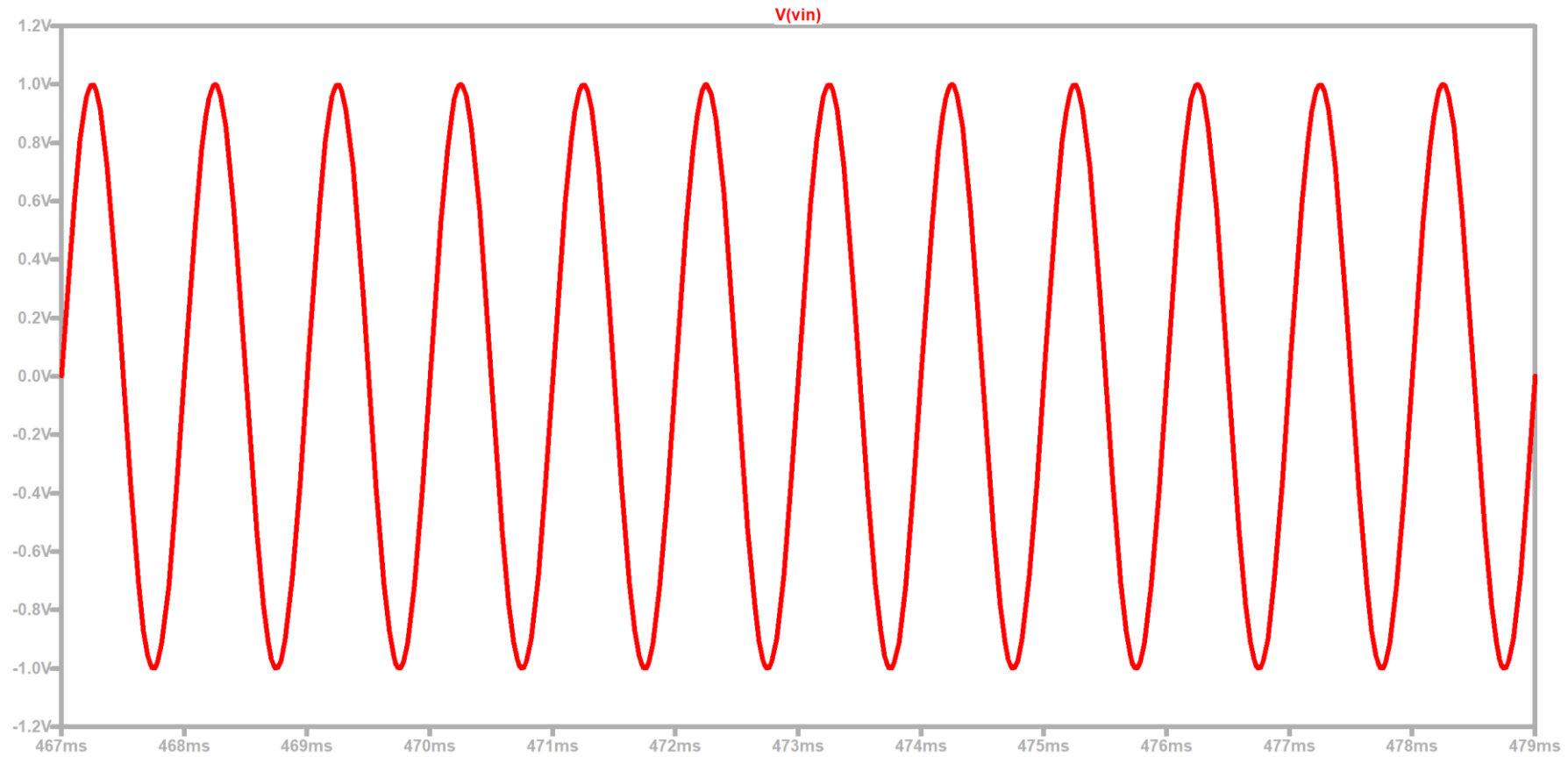
DIODE CIRCUITS

Diode Voltage Doubler



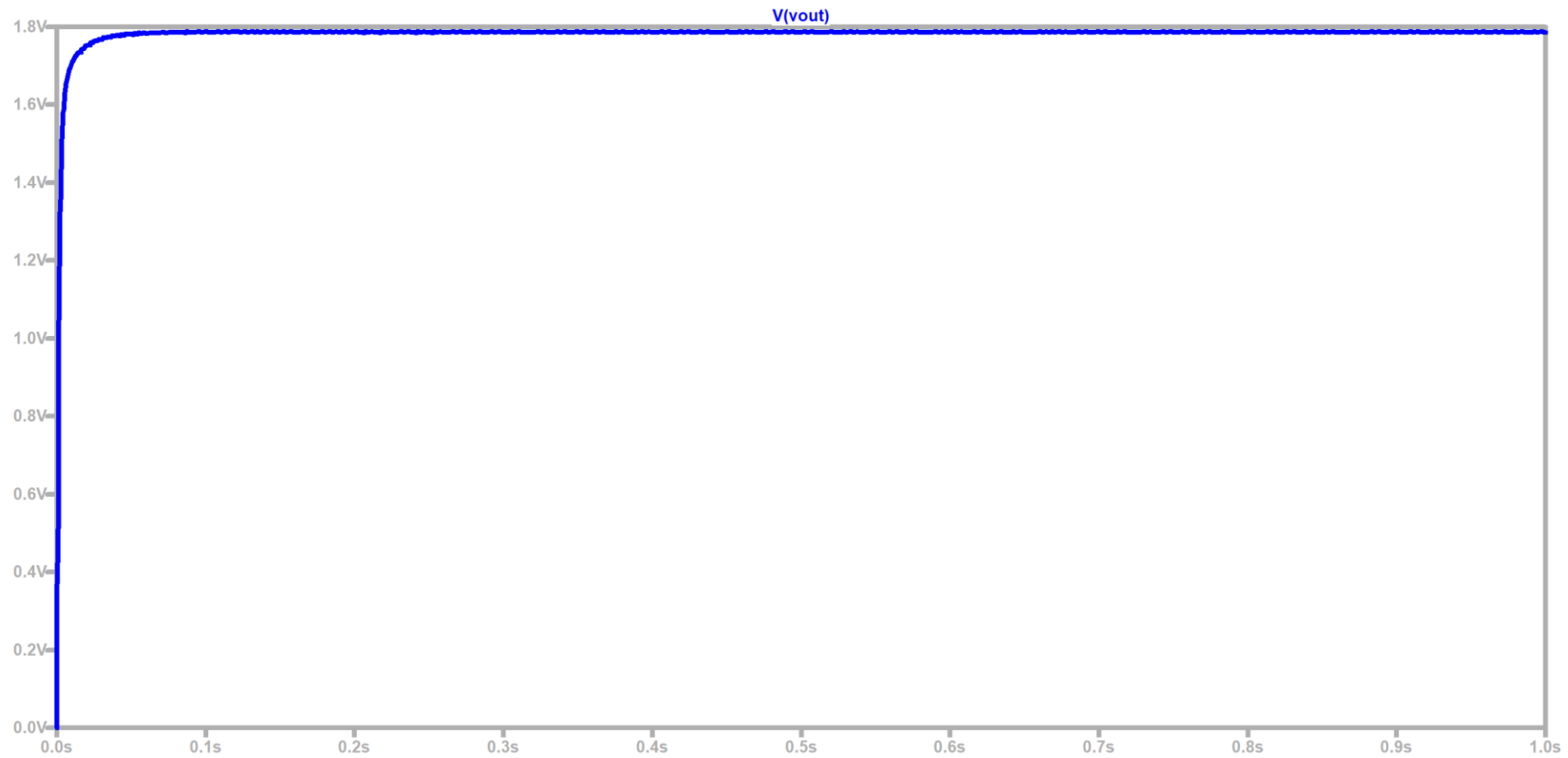
DIODE CIRCUITS

Diode Voltage Doubler



DIODE CIRCUITS

Diode Voltage Doubler



References

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- “AN11158, Understanding power MOSFET data sheet parameters”, NXP Application Note, Feb 2014
- Thermal Resistance Theory and Practice, Special Subject Book, January 2000, Infineon Technologies